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(54) **TISSUE VISUALIZATION AND
MANIPULATION SYSTEM**

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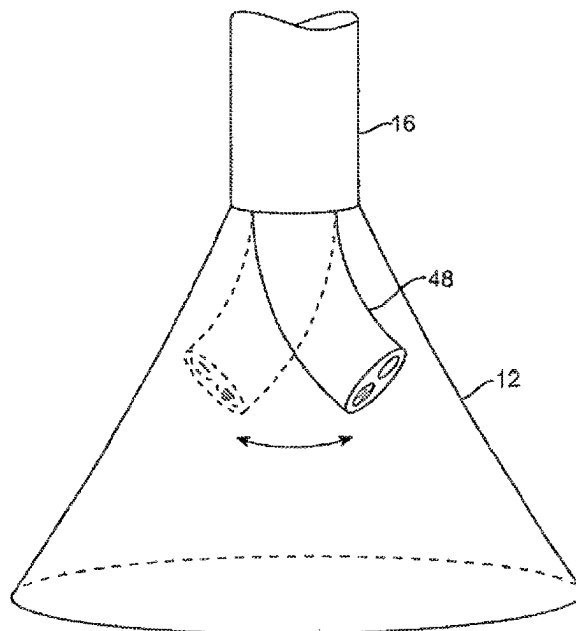
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(57) **ABSTRACT**

A tissue imaging assembly includes a steerable sheath and a
steerable catheter deployable within the sheath. The tissue
imaging assembly also includes an imaging hood at a distal
end of the steerable catheter. The imaging hood is independ-
ently translatable with respect to the steerable sheath and
the steerable catheter. The steerable catheter includes an
imaging element and a lumen for delivering an elongate
instrument out a distal region of the steerable catheter.

9 Claims, 43 Drawing Sheets



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No. 14/662,960, filed on Mar. 19, 2015, now abandoned, which is a continuation of application No. 12/947,198, filed on Nov. 16, 2010, now abandoned, which is a continuation of application No. 11/259,498, filed on Oct. 25, 2005, now Pat. No. 7,860,555.					
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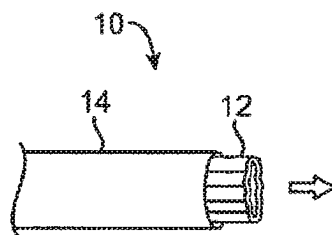


FIG. 1A

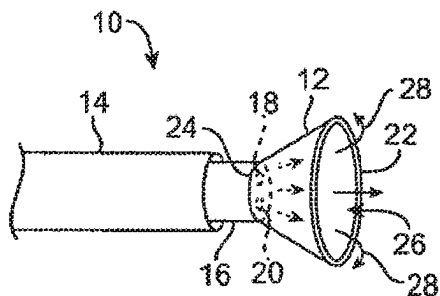


FIG. 1B

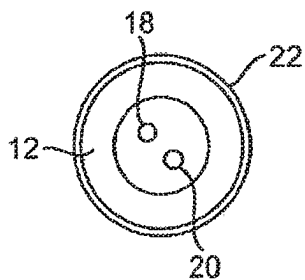


FIG. 1C

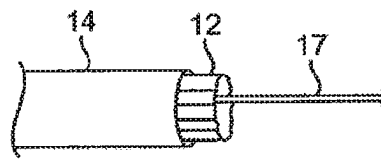


FIG. 1D

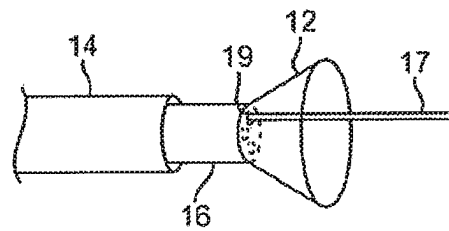


FIG. 1E

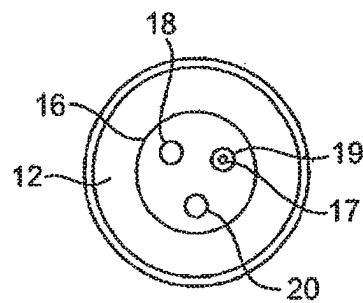


FIG. 1F

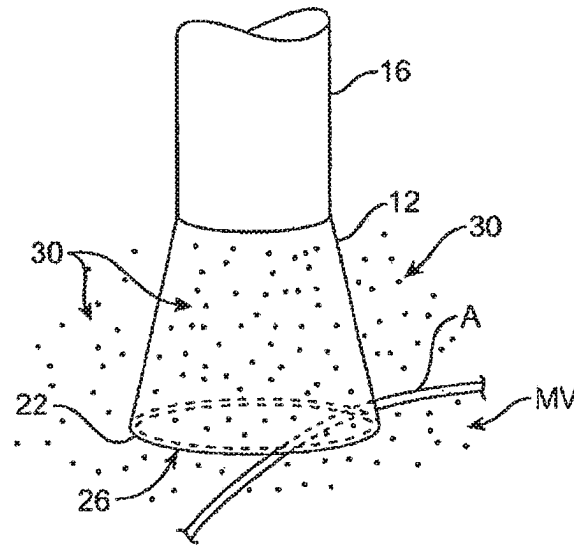


FIG. 2A

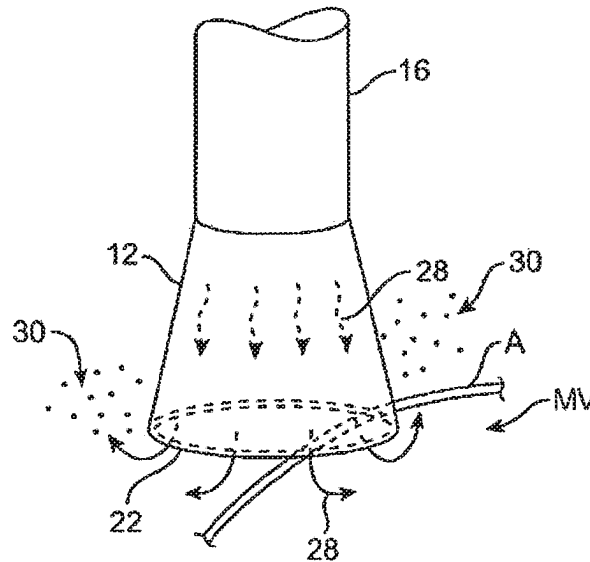


FIG. 2B

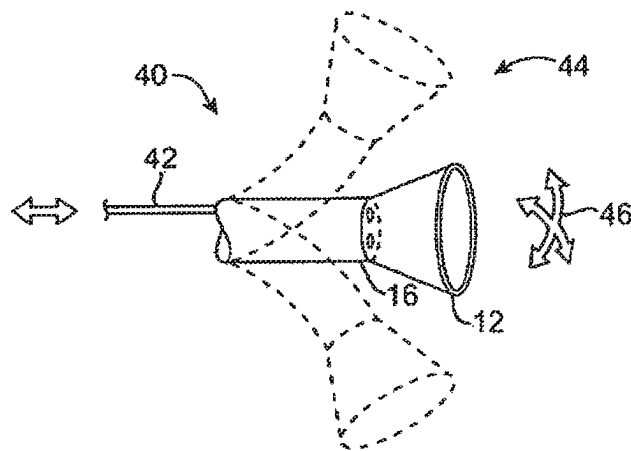


FIG. 3A

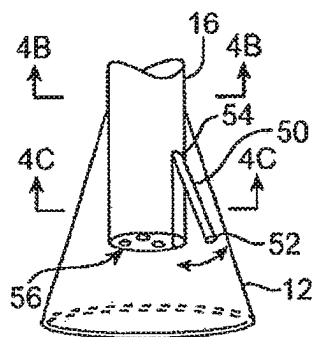


FIG. 4A

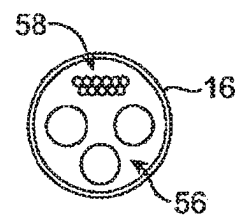


FIG. 4B

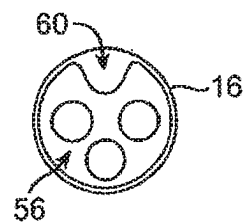


FIG. 4C

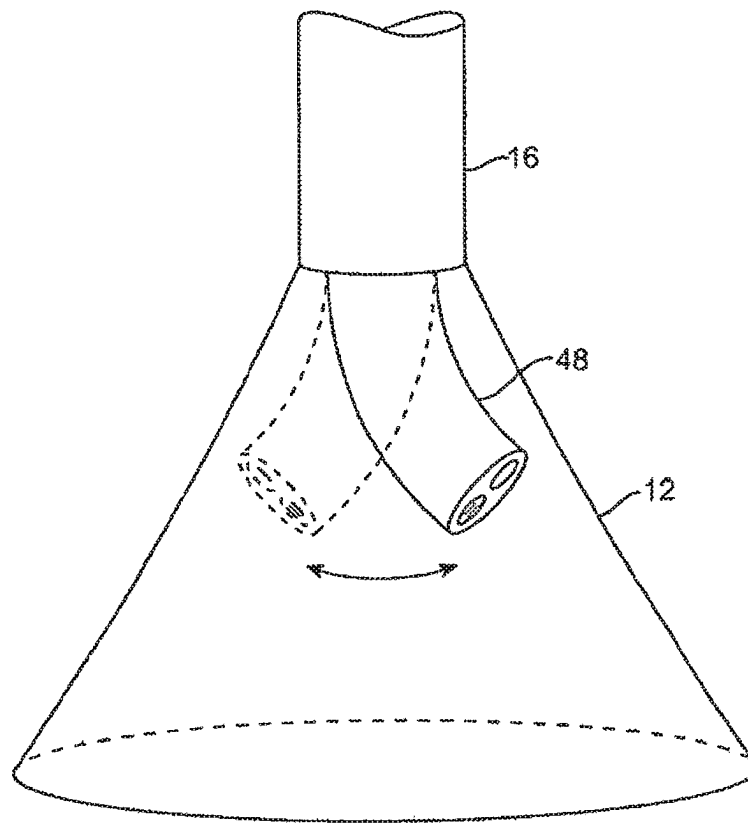


FIG. 3B

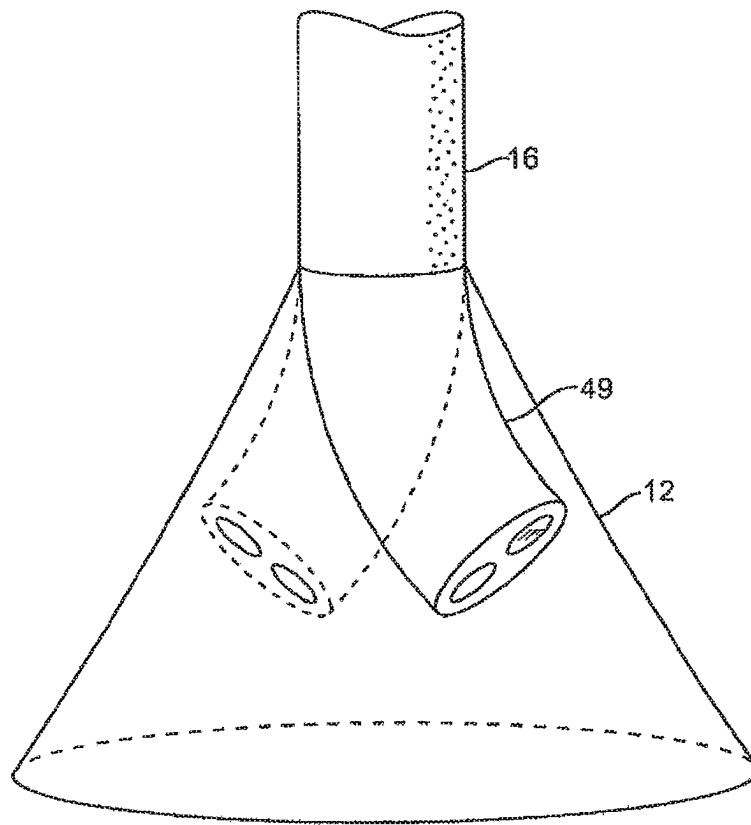


FIG. 3C

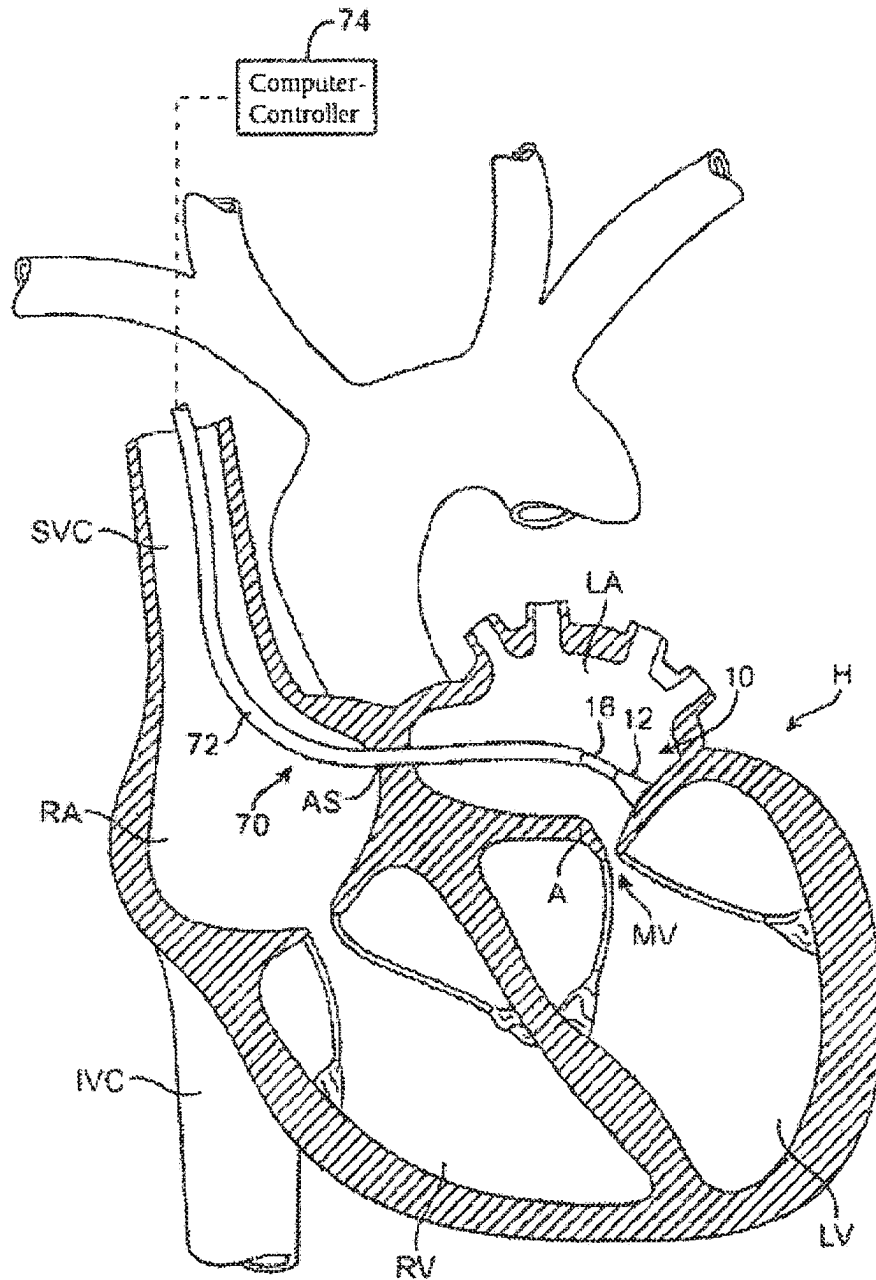


FIG. 5

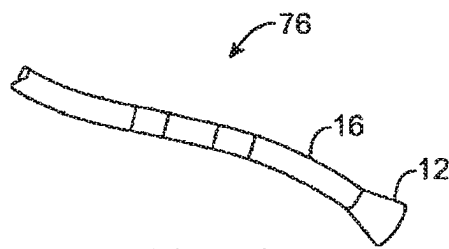


FIG. 6A

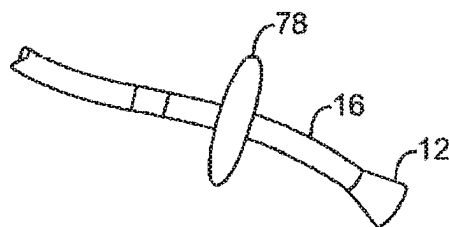


FIG. 6B

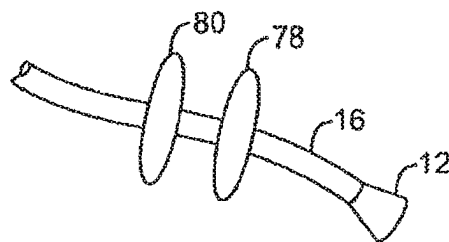


FIG. 6C

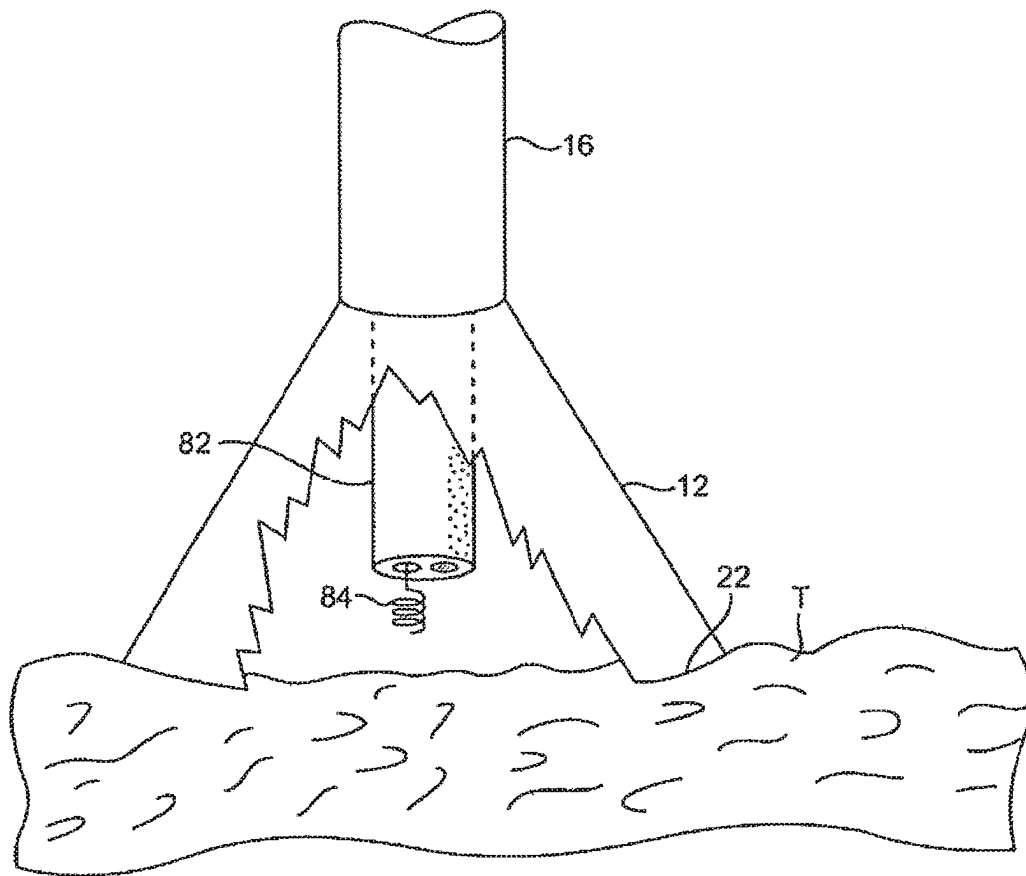


FIG. 7A

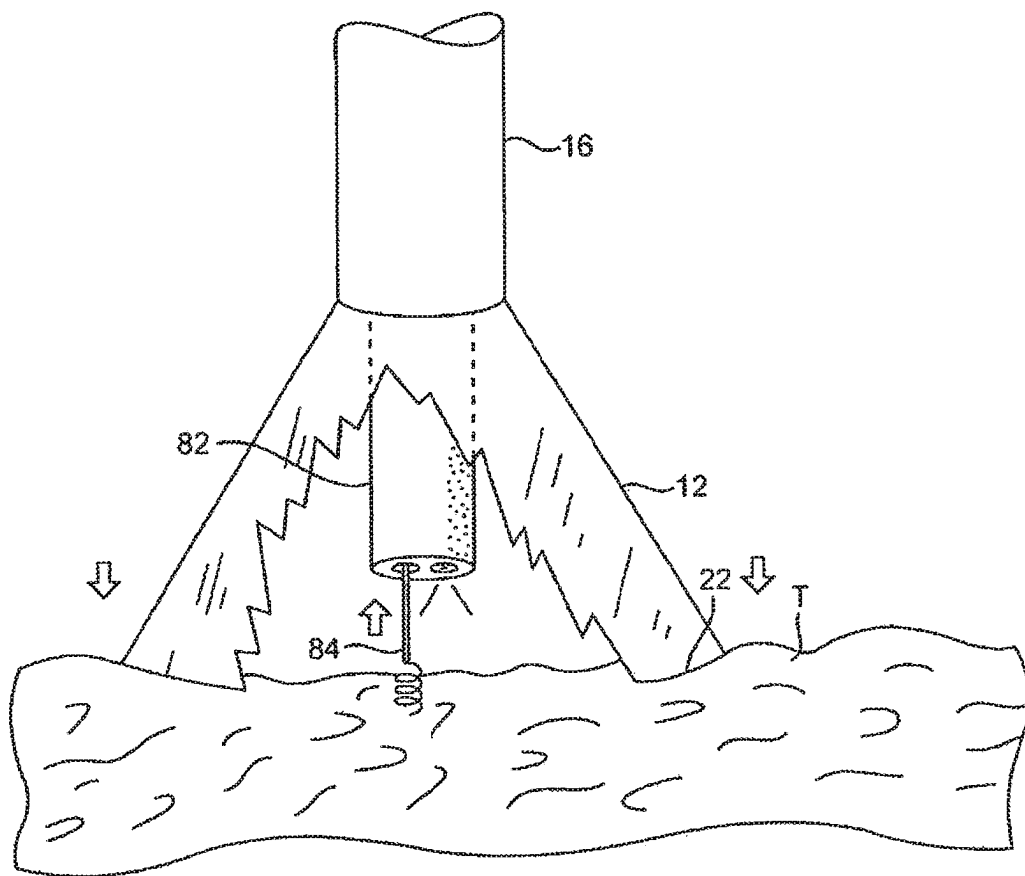


FIG. 7B

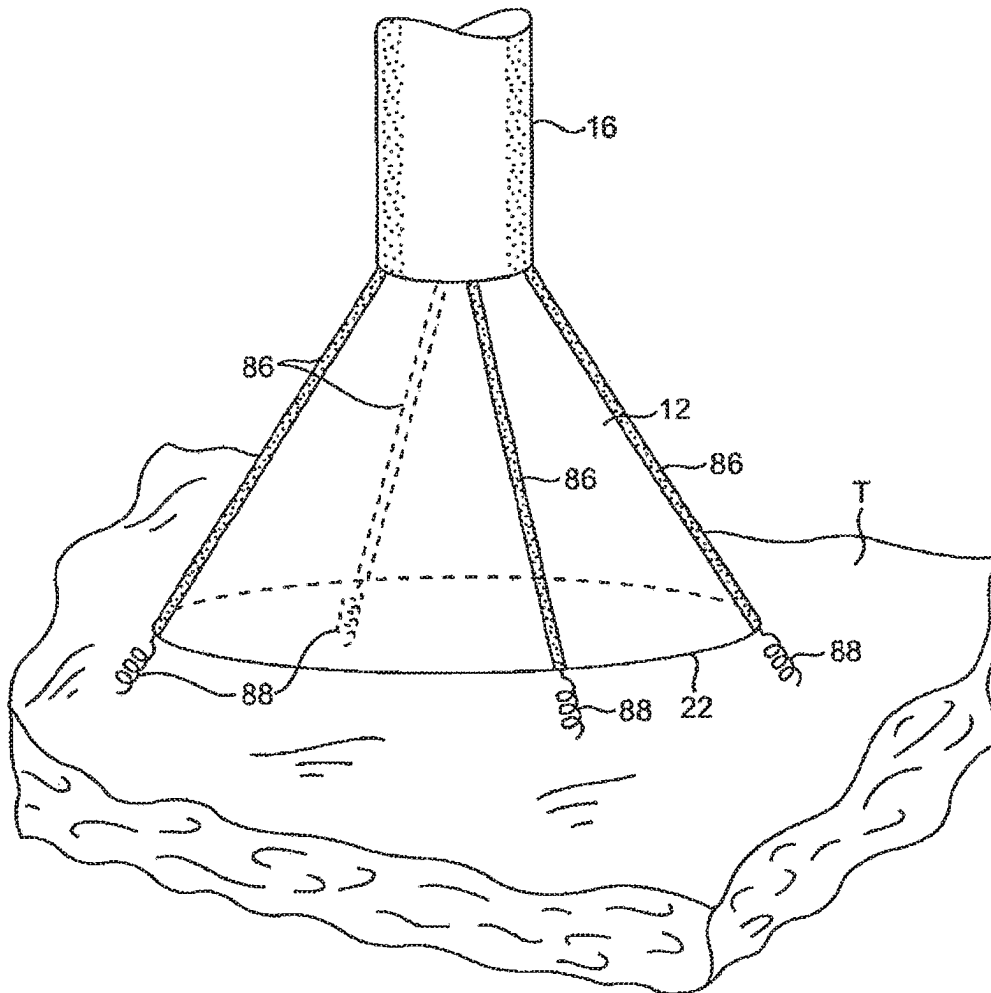


FIG. 7C

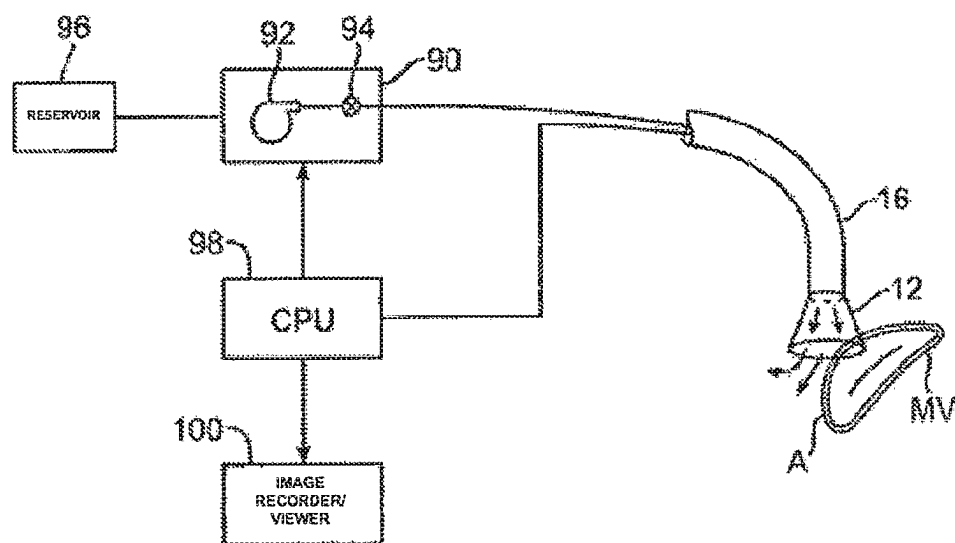


FIG. 8A

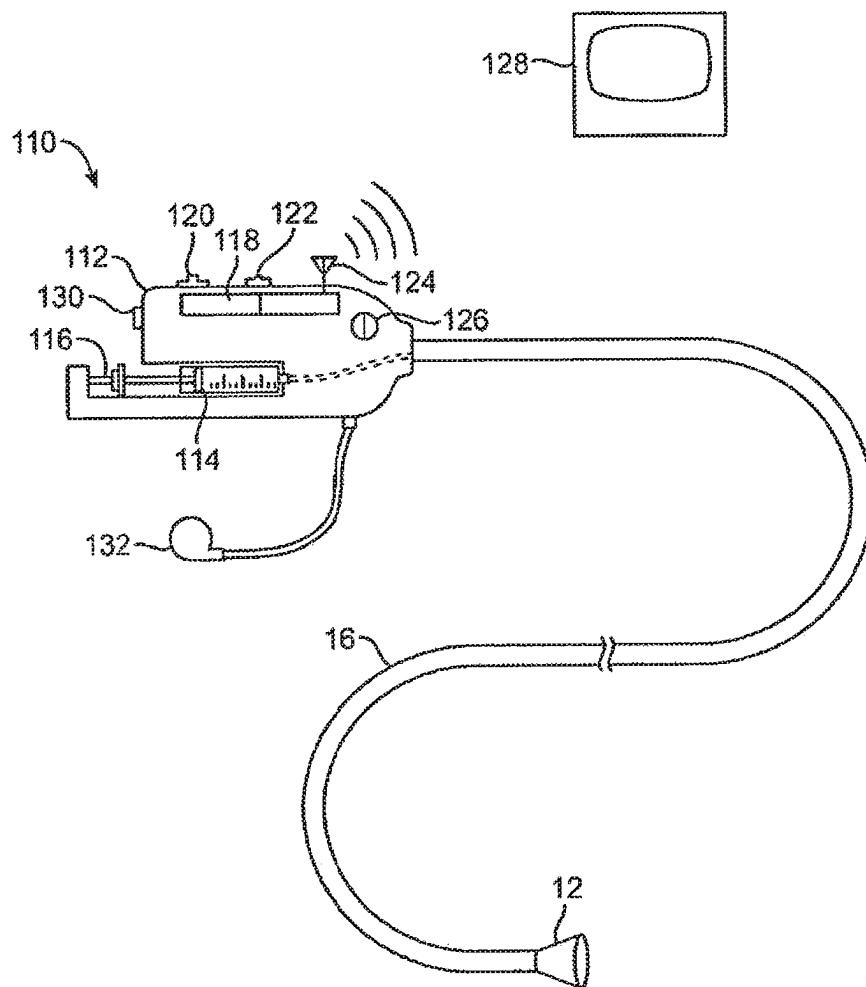


FIG. 8B

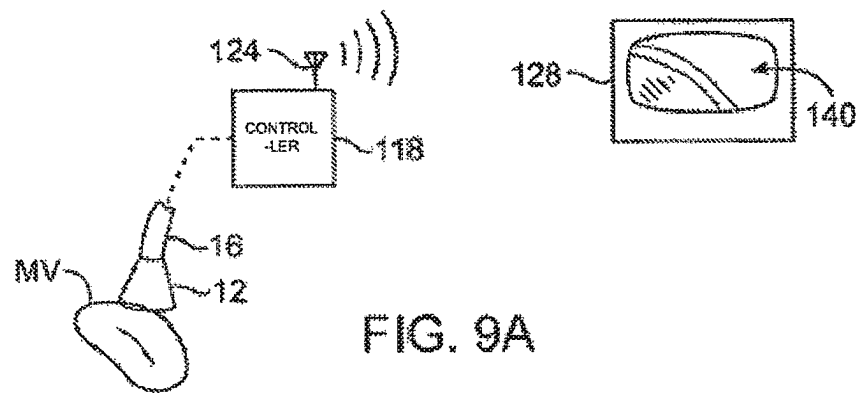


FIG. 9A

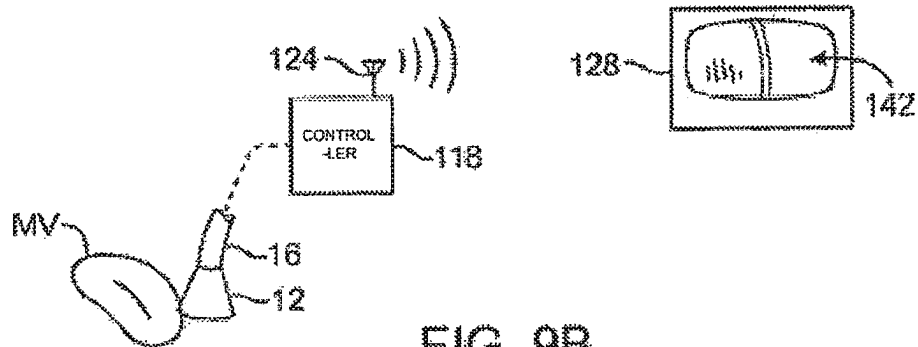


FIG. 99

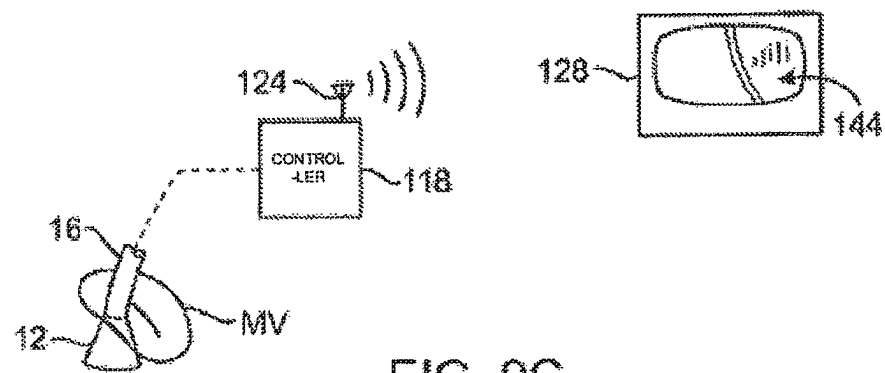


FIG. 9C

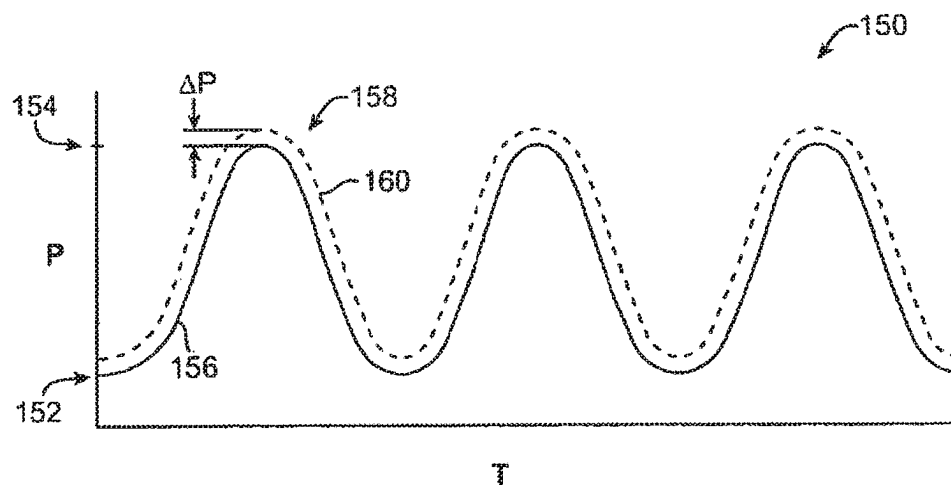


FIG. 10A

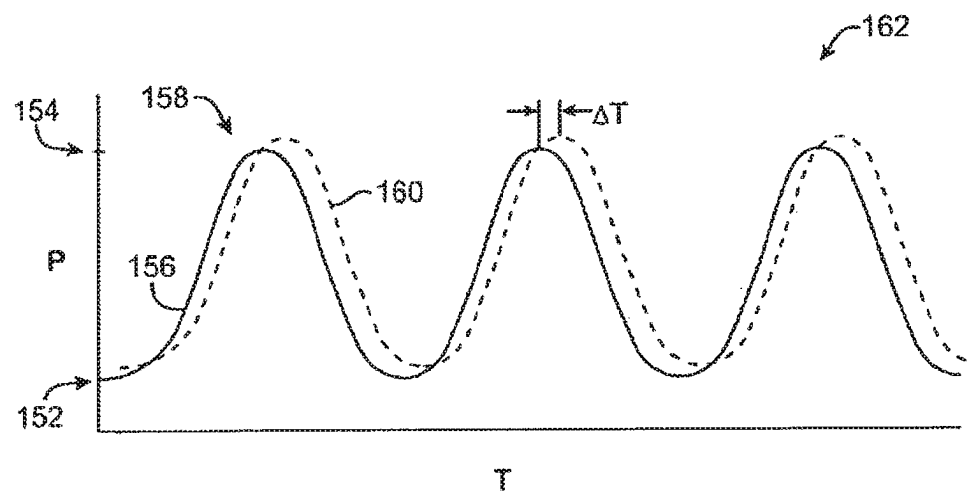


FIG. 10B

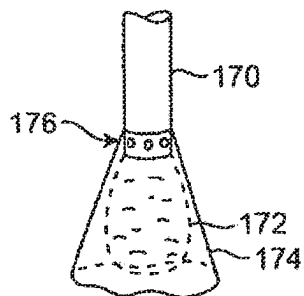


FIG. 11A

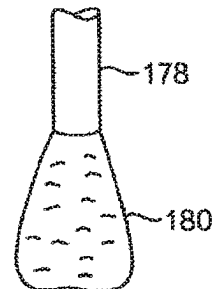


FIG. 11B

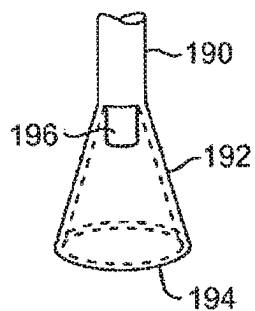


FIG. 13A

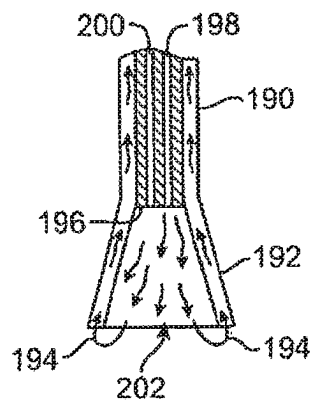


FIG. 13B

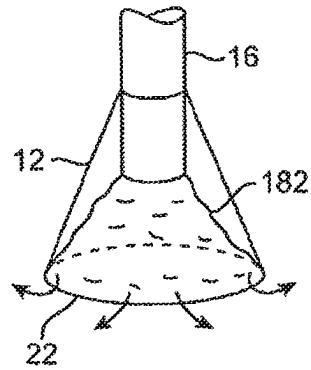


FIG. 12A

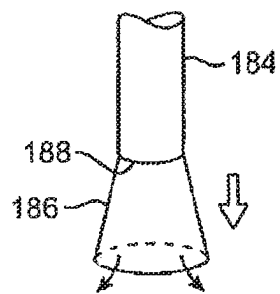


FIG. 12B

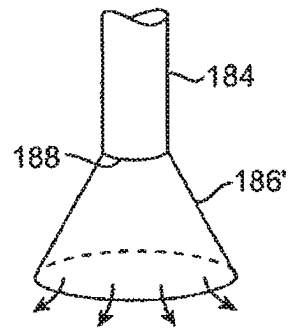


FIG. 12C

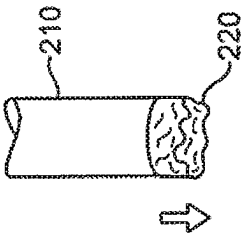


FIG. 14D

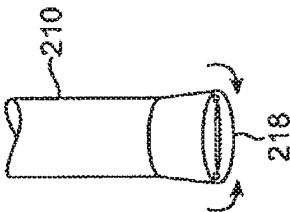


FIG. 14C

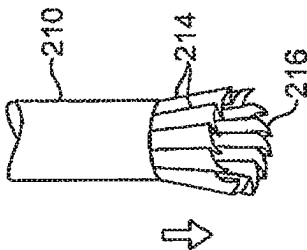


FIG. 14B

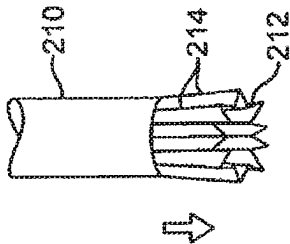


FIG. 14A

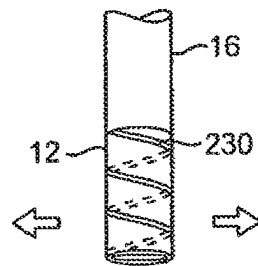


FIG. 15A

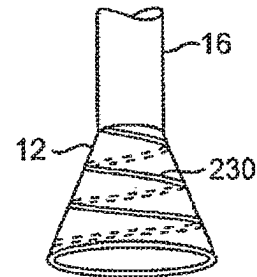


FIG. 15B

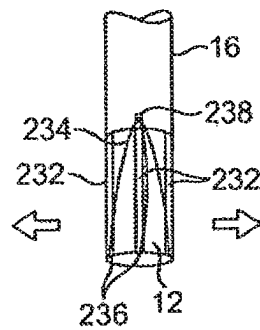


FIG. 16A

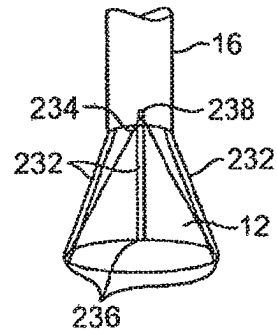


FIG. 16B

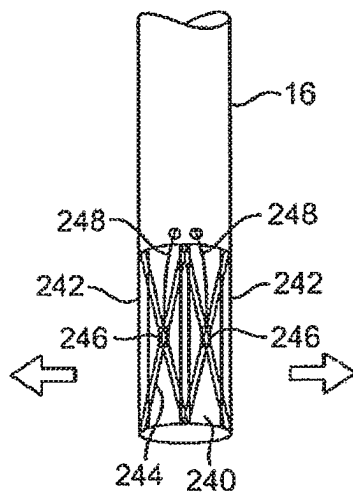


FIG. 17A

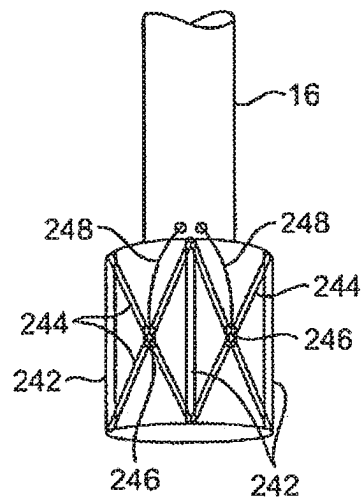


FIG. 17B

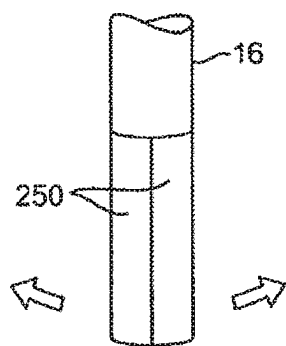


FIG. 18A

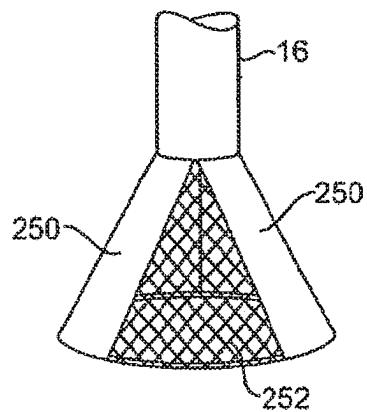


FIG. 18B

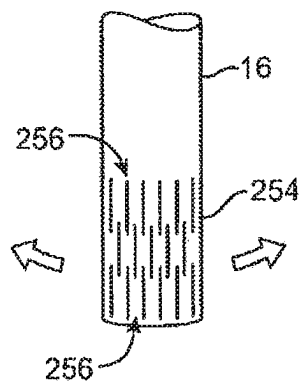


FIG. 19A

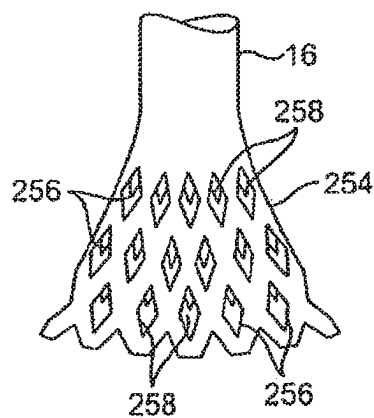


FIG. 19B

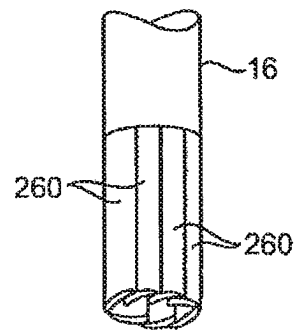


FIG. 20A

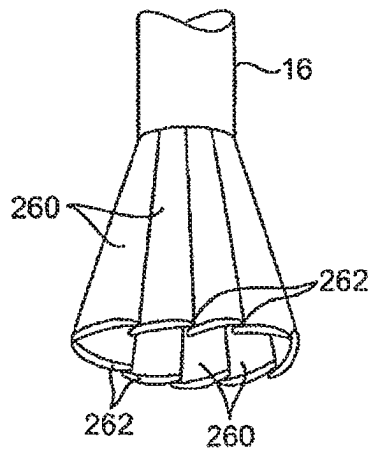


FIG. 20B

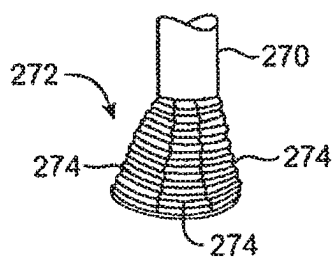


FIG. 21A

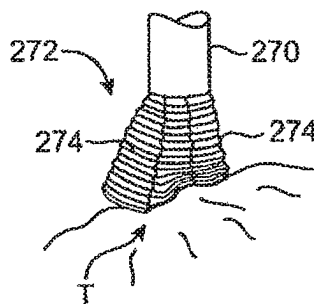


FIG. 21B

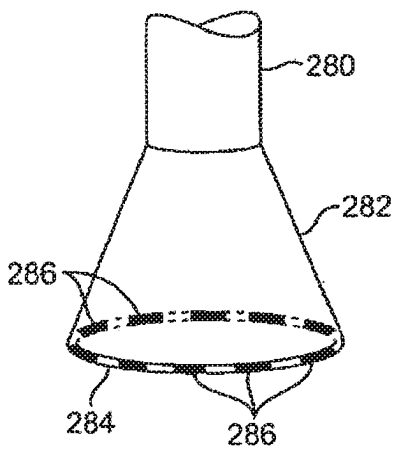


FIG. 22A

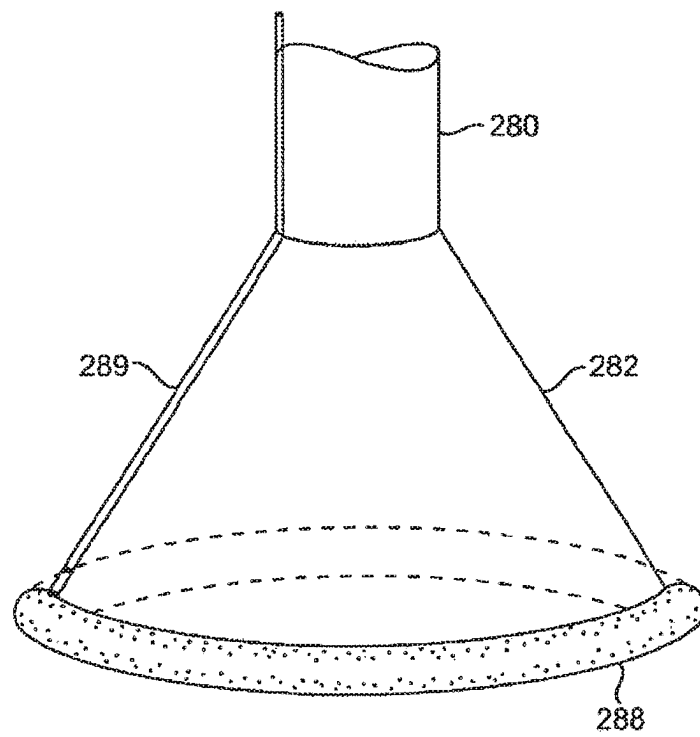


FIG. 22B

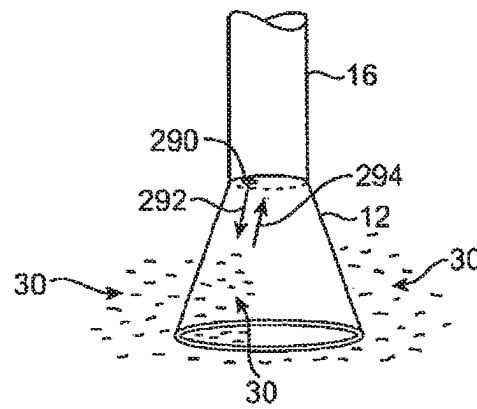


FIG. 23

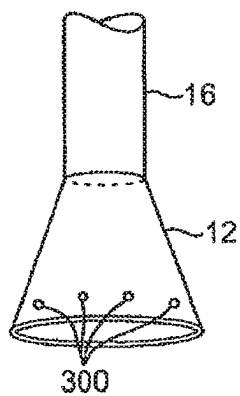


FIG. 24A

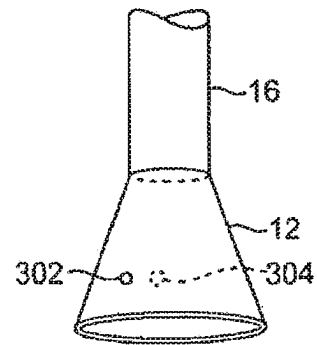


FIG. 24B

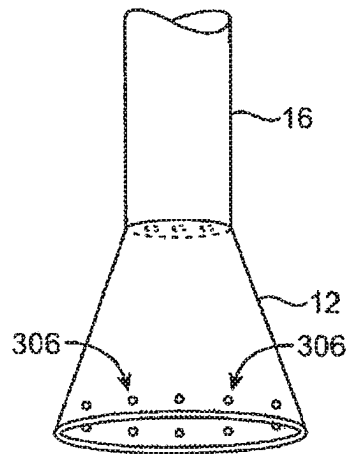


FIG. 25A

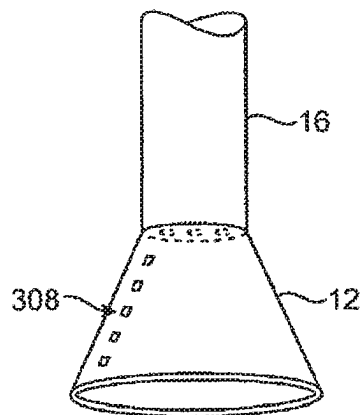


FIG. 25B

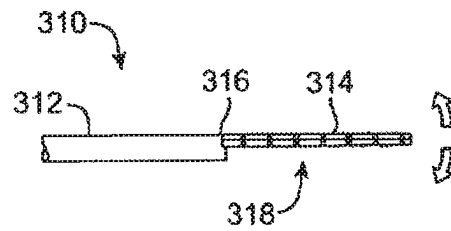


FIG. 26A

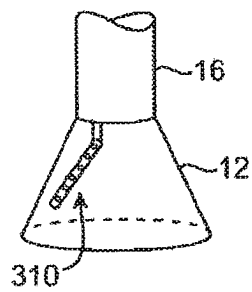


FIG. 26B

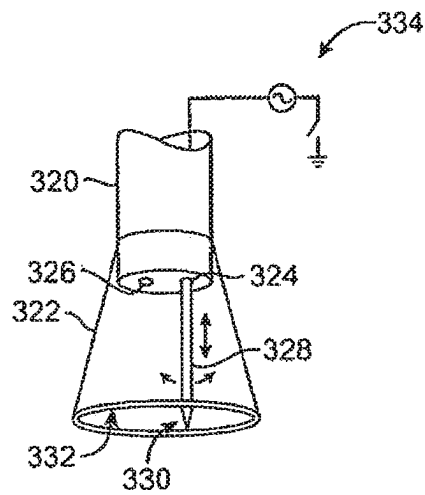


FIG. 27

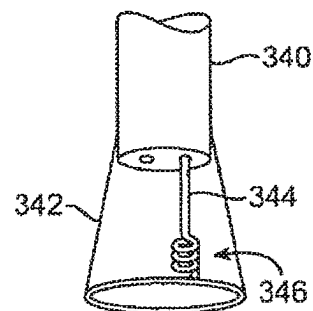


FIG. 28

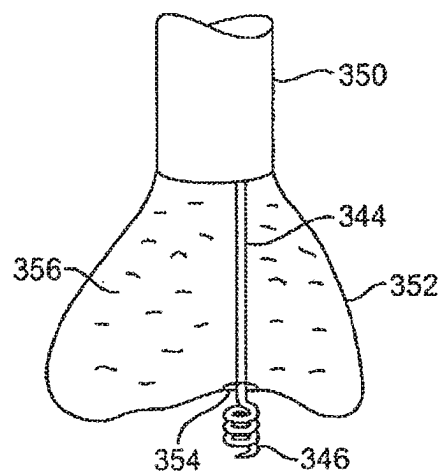


FIG. 29

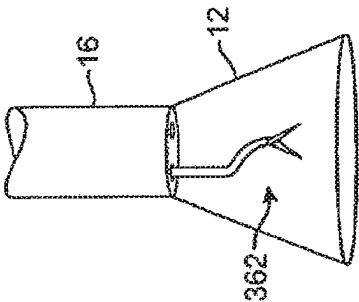


FIG. 30B

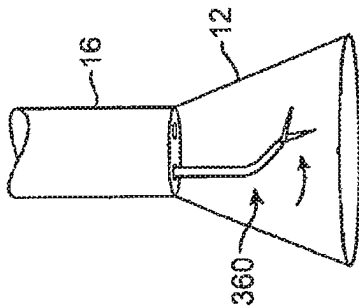


FIG. 30A

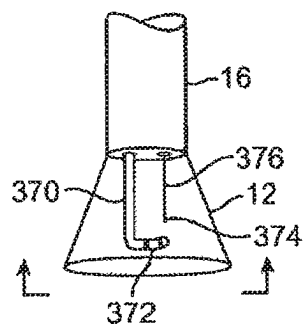


FIG. 31A

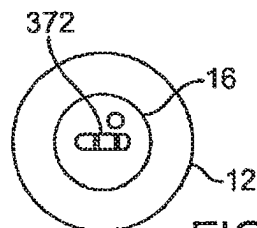


FIG. 31B

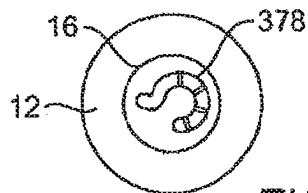


FIG. 31C

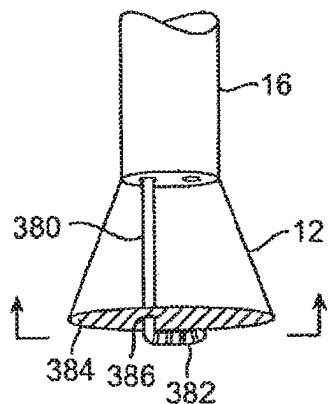


FIG. 32A

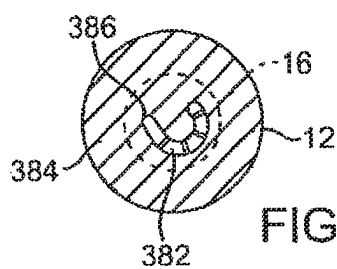


FIG. 32B

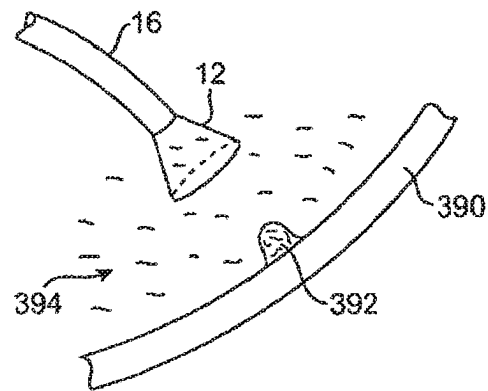


FIG. 33A

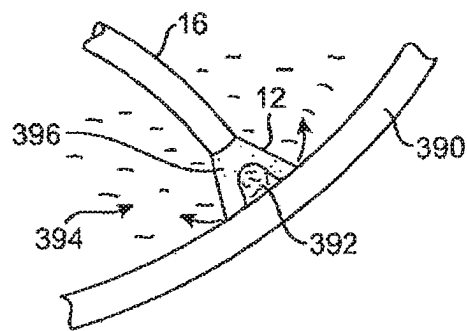


FIG. 33B

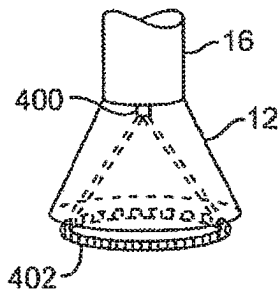


FIG 34A

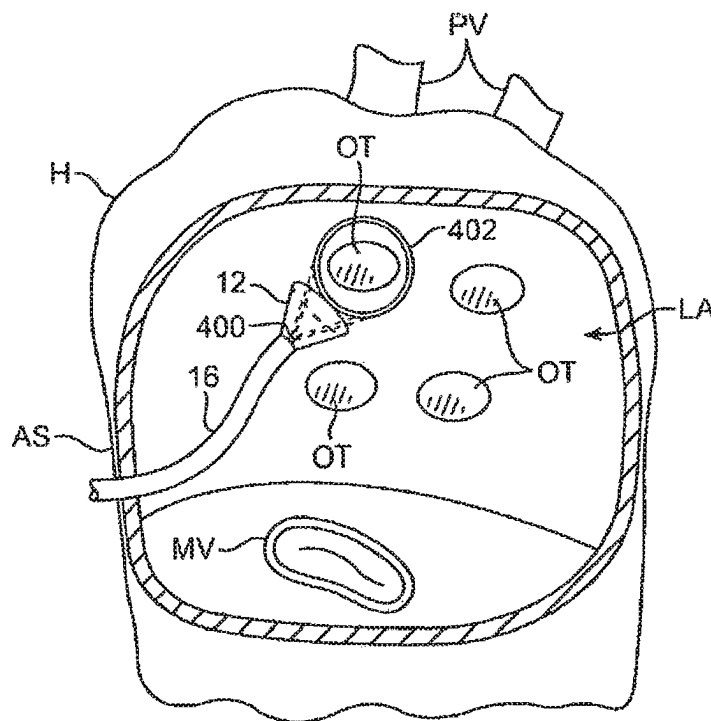


FIG. 34B

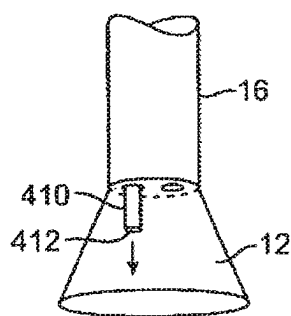


FIG. 35A

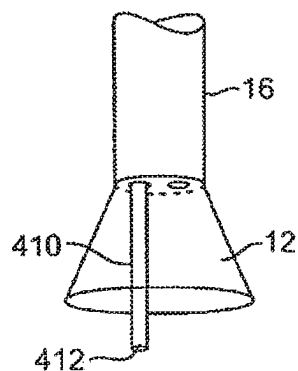


FIG. 35B

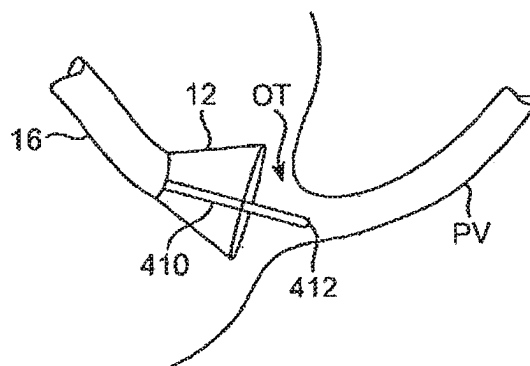


FIG. 35C

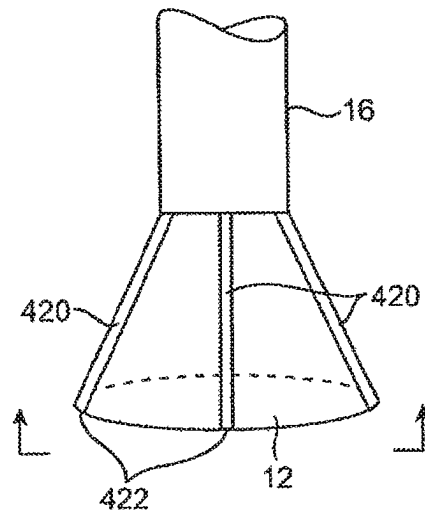


FIG. 36A

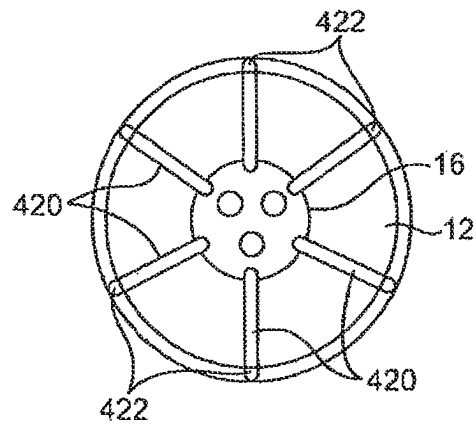


FIG. 36B

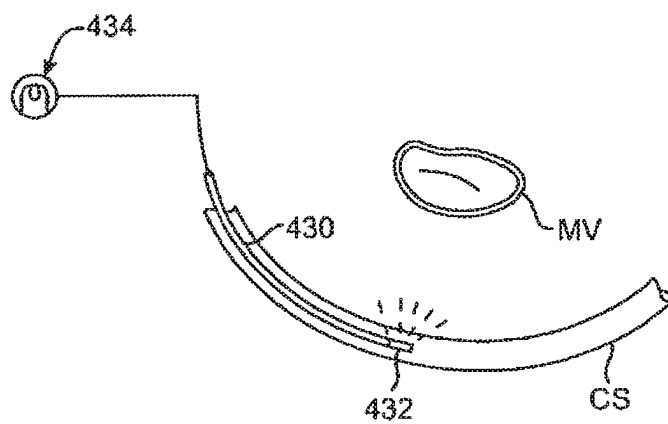


FIG. 37A

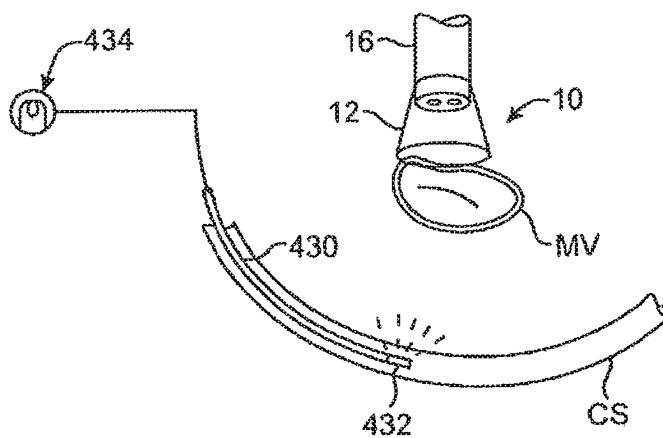


FIG. 37B

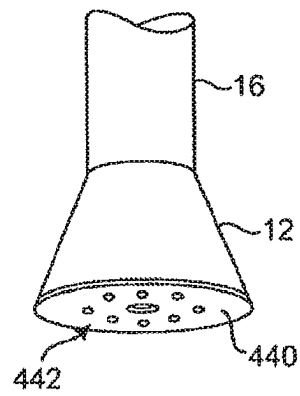


FIG. 38A

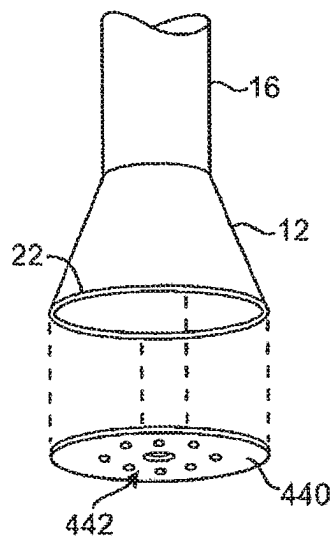


FIG. 38B

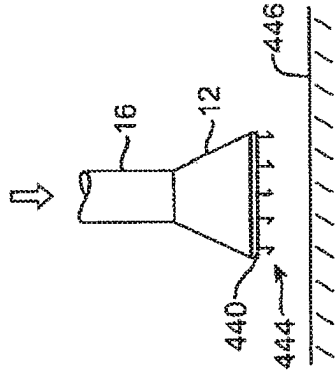


FIG. 39A

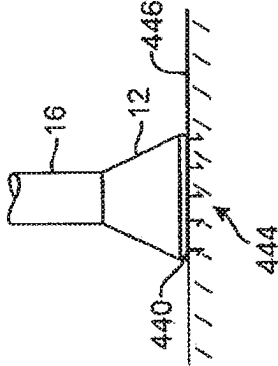


FIG. 39B

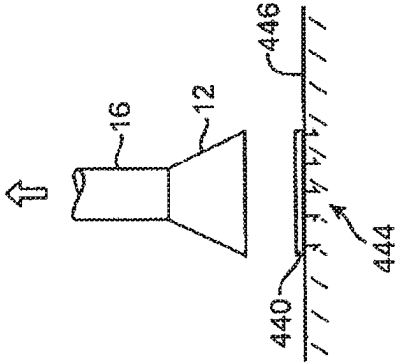


FIG. 39C

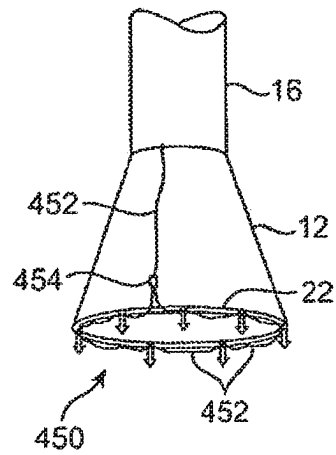


FIG. 40A

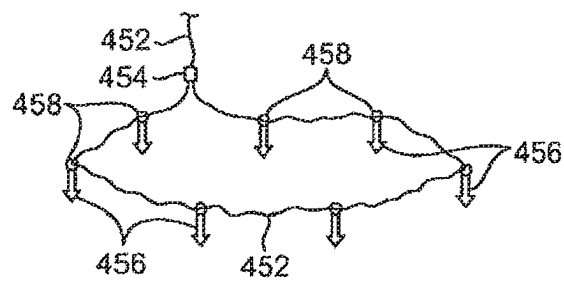


FIG. 40B

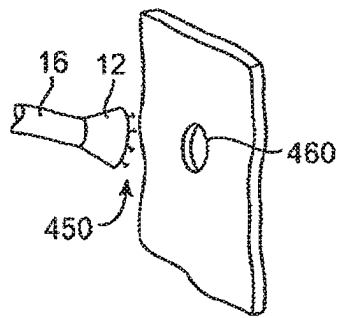


FIG. 41A

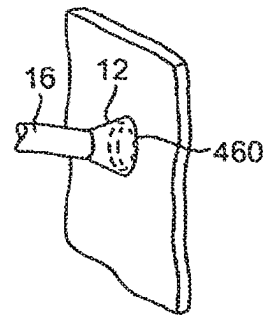


FIG. 41B

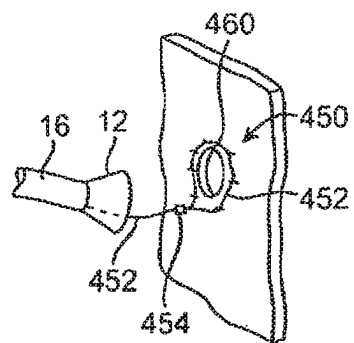


FIG. 41C

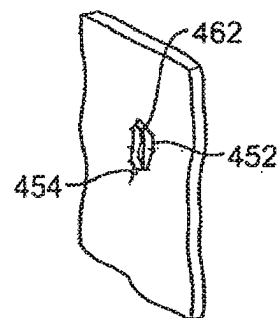


FIG. 41D

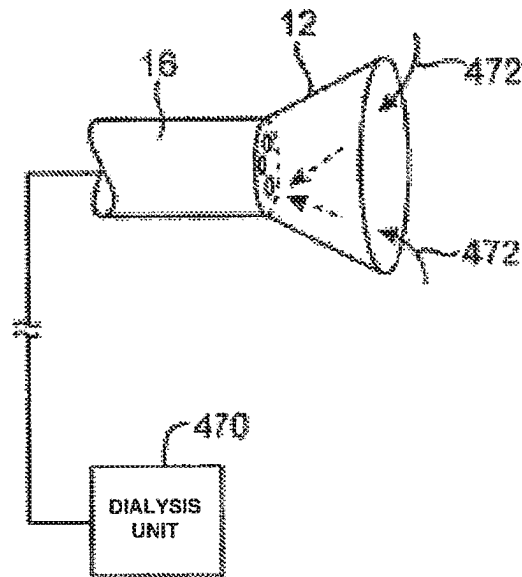


FIG. 42

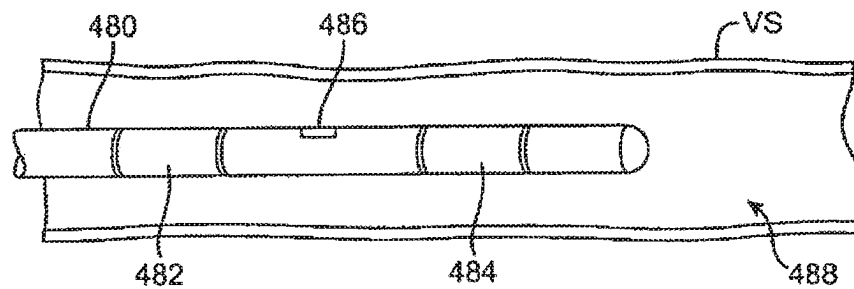


FIG. 43A

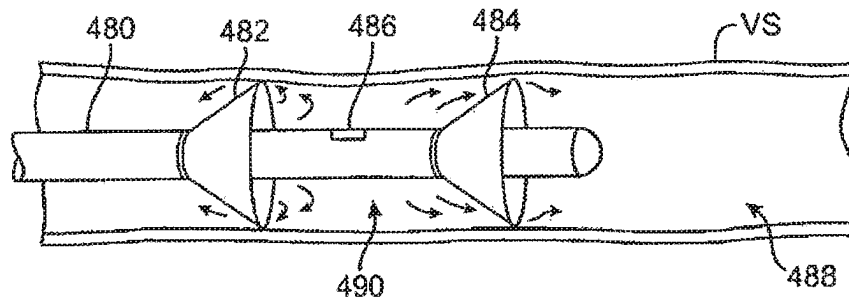


FIG. 43B

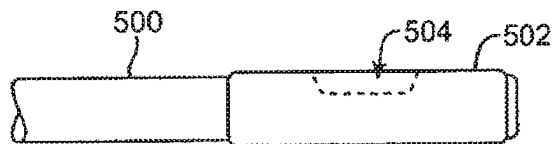


FIG. 44A



FIG. 44B

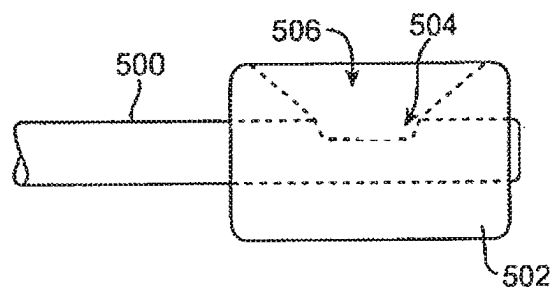


FIG. 45A

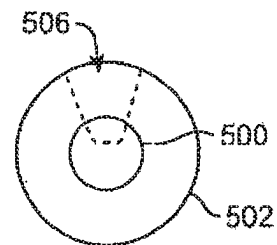


FIG. 45C

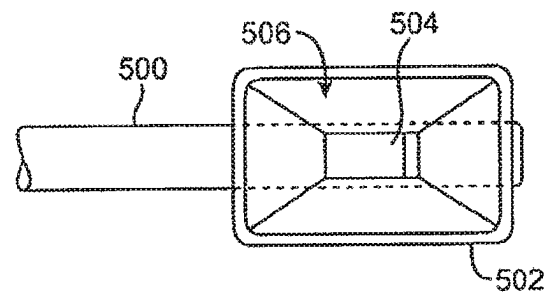


FIG. 45B

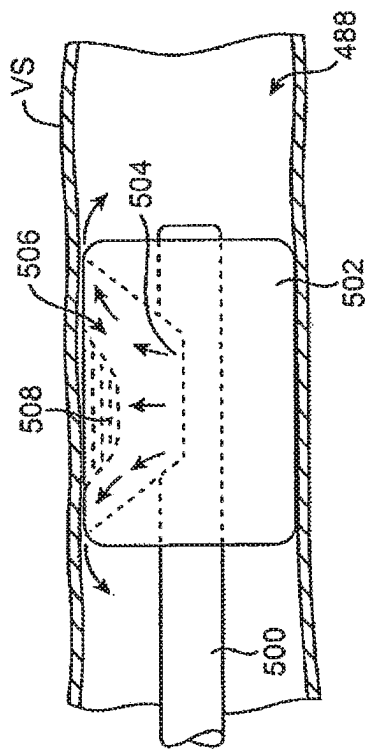


FIG. 46A

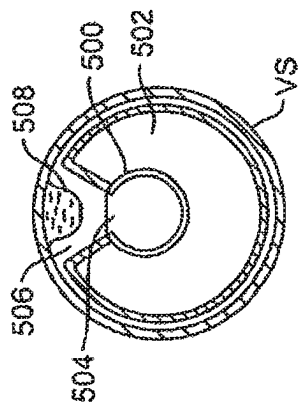


FIG. 46B

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TISSUE VISUALIZATION AND MANIPULATION SYSTEM

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 15/999,543 filed Aug. 20, 2018 which is a continuation of U.S. patent application Ser. No. 14/662,960 filed Mar. 19, 2015 which is a continuation of U.S. patent application Ser. No. 12/947,198 filed Nov. 16, 2010, which is a continuation of U.S. patent application Ser. No. 11/259,498 filed Oct. 25, 2005, which claims the benefit of priority to U.S. Provisional Patent Application No. 60/649,246 filed Feb. 2, 2005, the disclosures of which are hereby incorporated by reference in their entirety.

FIELD OF THE INVENTION

The present invention relates generally to medical devices used for visualizing and/or manipulating regions of tissue within a body. More particularly, the present invention relates to apparatus and methods for visualizing and/or manipulating tissue regions within a body lumen, e.g., tissue surrounding or adjacent to valves within a heart, which are generally difficult to image because of surrounding opaque bodily fluids such as blood or the tissue of the inter-atrial septum for trans-septal procedures.

Conventional devices for visualizing interior regions of a body lumen are known. For example, ultrasound devices have been used to produce images from within a body in vivo. Ultrasound has been used both with and without contrast agents, which typically enhance ultrasound-derived images.

Other conventional methods have utilized catheters or probes having position sensors deployed within the body lumen, such as the interior of a cardiac chamber. These types of positional sensors are typically used to determine the movement of a cardiac tissue surface or the electrical activity within the cardiac tissue. When a sufficient number of points have been sampled by the sensors, a “map” of the cardiac tissue may be generated.

Another conventional device utilizes an inflatable balloon which is typically introduced intravascularly in a deflated state and then inflated against the tissue region to be examined. Imaging is typically accomplished by an optical fiber or other apparatus such as electronic chips for viewing the tissue through the membrane(s) of the inflated balloon. Moreover, the balloon must generally be inflated for imaging. Other conventional balloons utilize a cavity or depression formed at a distal end of the inflated balloon. This cavity or depression is pressed against the tissue to be examined and is flushed with a clear fluid to provide a clear pathway through the blood.

However, such imaging balloons have many inherent disadvantages. For instance, such balloons generally require that the balloon be inflated to a relatively large size which may undesirably displace surrounding tissue and interfere with fine positioning of the imaging system against the tissue. Moreover, the working area created by such inflatable balloons are generally cramped and limited in size. Furthermore, inflated balloons may be susceptible to pressure changes in the surrounding fluid. For example, if the environment surrounding the inflated balloon undergoes pressure changes, e.g., during systolic and diastolic pressure cycles in a beating heart, the constant pressure change may

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affect the inflated balloon volume and its positioning to produce unsteady or undesirable conditions for optimal tissue imaging.

Accordingly, these types of imaging modalities are generally unable to provide desirable images useful for sufficient diagnosis and therapy of the endoluminal structure, due in part to factors such as dynamic forces generated by the natural movement of the heart. Moreover, anatomic structures within the body can occlude or obstruct the image acquisition process. Also, the presence and movement of opaque bodily fluids such as blood generally make in vivo imaging of tissue regions within the heart difficult.

Other external imaging modalities are also conventionally utilized. For example, computed tomography (CT) and magnetic resonance imaging (MRI) are typical modalities which are widely used to obtain images of body lumens such as the interior chambers of the heart. However, such imaging modalities fail to provide real-time imaging for intra-operative therapeutic procedures. Fluoroscopic imaging, for instance, is widely used to identify anatomic landmarks within the heart and other regions of the body. However, fluoroscopy fails to provide an accurate image of the tissue quality or surface and also fails to provide for instrumentation for performing tissue manipulation or other therapeutic procedures upon the visualized tissue regions. In addition, fluoroscopy provides a shadow of the intervening tissue onto a plate or sensor when it may be desirable to view the intraluminal surface of the tissue to diagnose pathologies or to perform some form of therapy on it.

Thus, a tissue imaging system which is able to provide real-time in vivo images of tissue regions within body lumens such as the heart through opaque media such as blood and which also provide instruments for therapeutic procedures upon the visualized tissue are desirable.

BRIEF SUMMARY OF THE INVENTION

A tissue imaging and manipulation apparatus that may be utilized for procedures within a body lumen, such as the heart, in which visualization of the surrounding tissue is made difficult, if not impossible, by medium contained within the lumen such as blood, is described below. Generally, such a tissue imaging and manipulation apparatus comprises an optional delivery catheter or sheath through which a deployment catheter and imaging hood may be advanced for placement against or adjacent to the tissue to be imaged.

The deployment catheter may define a fluid delivery lumen therethrough as well as an imaging lumen within which an optical imaging fiber or assembly may be disposed for imaging tissue. When deployed, the imaging hood may be expanded into any number of shapes, e.g., cylindrical, conical as shown, semi-spherical, etc., provided that an open area or field is defined by the imaging hood. The open area is the area within which the tissue region of interest may be imaged. The imaging hood may also define an atraumatic contact lip or edge for placement or abutment against the tissue region of interest. Moreover, the distal end of the deployment catheter or separate manipulatable catheters may be articulated through various controlling mechanisms such as push-pull wires manually or via computer control.

The deployment catheter may also be stabilized relative to the tissue surface through various methods. For instance, inflatable stabilizing balloons positioned along a length of the catheter may be utilized, or tissue engagement anchors may be passed through or along the deployment catheter for temporary engagement of the underlying tissue.

In operation, after the imaging hood has been deployed, fluid may be pumped at a positive pressure through the fluid delivery lumen until the fluid fills the open area completely and displaces any blood from within the open area. The fluid may comprise any biocompatible fluid, e.g., saline, water, plasma, Fluorinert™, etc., which is sufficiently transparent to allow for relatively undistorted visualization through the fluid. The fluid may be pumped continuously or intermittently to allow for image capture by an optional processor which may be in communication with the assembly.

The imaging hood may be formed into any number of configurations and the imaging assembly may also be utilized with any number of therapeutic tools which may be deployed through the deployment catheter.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1A shows a side view of one variation of a tissue imaging apparatus during deployment from a sheath or delivery catheter.

FIG. 1B shows the deployed tissue imaging apparatus of FIG. 1A having an optionally expandable hood or sheath attached to an imaging and/or diagnostic catheter.

FIG. 1C shows an end view of a deployed imaging apparatus.

FIGS. 1D to 1F show the apparatus of FIGS. 1A to 1C with an additional lumen, e.g., for passage of a guidewire therethrough.

FIGS. 2A and 2B show one example of a deployed tissue imager positioned against or adjacent to the tissue to be imaged and a flow of fluid, such as saline, displacing blood from within the expandable hood.

FIG. 3A shows an articulatable imaging assembly which may be manipulated via push-pull wires or by computer control.

FIGS. 3B and 3C show steerable instruments, respectively, where an articulatable delivery catheter may be steered within the imaging hood or a distal portion of the deployment catheter itself may be steered.

FIGS. 4A to 4C show side and cross-sectional end views, respectively, of another variation having an off-axis imaging capability.

FIG. 5 shows an illustrative view of an example of a tissue imager advanced intravascularly within a heart for imaging tissue regions within an atrial chamber.

FIGS. 6A to 6C illustrate deployment catheters having one or more optional inflatable balloons or anchors for stabilizing the device during a procedure.

FIGS. 7A and 7B illustrate a variation of an anchoring mechanism such as a helical tissue piercing device for temporarily stabilizing the imaging hood relative to a tissue surface.

FIG. 7C shows another variation for anchoring the imaging hood having one or more tubular support members integrated with the imaging hood, each support members may define a lumen therethrough for advancing a helical tissue anchor within.

FIG. 8A shows an illustrative example of one variation of how a tissue imager may be utilized with an imaging device.

FIG. 8B shows a further illustration of a hand-held variation of the fluid delivery and tissue manipulation system.

FIGS. 9A to 9C illustrate an example of capturing several images of the tissue at multiple regions.

FIGS. 10A and 10B show charts illustrating how fluid pressure within the imaging hood may be coordinated with the surrounding blood pressure; the fluid pressure in the

imaging hood may be coordinated with the blood pressure or it may be regulated based upon pressure feedback from the blood.

FIG. 11A shows a side view of another variation of a tissue imager having an imaging balloon within an expandable hood.

FIG. 11B shows another variation of a tissue imager utilizing a translucent or transparent imaging balloon.

FIG. 12A shows another variation in which a flexible expandable or distensible membrane may be incorporated within the imaging hood to alter the volume of fluid dispensed.

FIGS. 12B and 12C show another variation in which the imaging hood may be partially or selectively deployed from the catheter to alter the area of the tissue being visualized as well as the volume of the dispensed fluid.

FIGS. 13A and 13B show exemplary side and cross-sectional views, respectively, of another variation in which the injected fluid may be drawn back into the device for minimizing fluid input into a body being treated.

FIGS. 14A to 14D show various configurations and methods for configuring an imaging hood into a low-profile for delivery and/or deployment.

FIGS. 15A and 15B show an imaging hood having an helically expanding frame or support.

FIGS. 16A and 16B show another imaging hood having one or more hood support members, which are pivotably attached at their proximal ends to deployment catheter, integrated with a hood membrane.

FIGS. 17A and 17B show yet another variation of the imaging hood having at least two or more longitudinally positioned support members supporting the imaging hood membrane where the support members are movable relative to one another via a torquing or pulling or pushing force.

FIGS. 18A and 18B show another variation where a distal portion of the deployment catheter may have several pivoting members which form a tubular shape in its low profile configuration.

FIGS. 19A and 19B show another variation where the distal portion of deployment catheter may be fabricated from a flexible metallic or polymeric material to form a radially expanding hood.

FIGS. 20A and 20B show another variation where the imaging hood may be formed from a plurality of overlapping hood members which overlies one another in an overlapping pattern.

FIGS. 21A and 21B show another example of an expandable hood which is highly conformable against tissue anatomy with varying geography.

FIG. 22A shows yet another example of an expandable hood having a number of optional electrodes placed about the contact edge or lip of the hood for sensing tissue contact or detecting arrhythmias.

FIG. 22B shows another variation for conforming the imaging hood against the underlying tissue where an inflatable contact edge may be disposed around the circumference of the imaging hood.

FIG. 23 shows a variation of the system which may be instrumented with a transducer for detecting the presence of blood seeping back into the imaging hood.

FIGS. 24A and 24B show variations of the imaging hood instrumented with sensors for detecting various physical parameters; the sensors may be instrumented around the outer surface of the imaging hood and also within the imaging hood.

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FIGS. 25A and 25B show a variation where the imaging hood may have one or more LEDs over the hood itself for providing illumination of the tissue to be visualized.

FIGS. 26A and 26B show another variation in which a separate illumination tool having one or more LEDs mounted thereon may be utilized within the imaging hood.

FIG. 27 shows one example of how a therapeutic tool may be advanced through the tissue imager for treating a tissue region of interest.

FIG. 28 shows another example of a helical therapeutic tool for treating the tissue region of interest.

FIG. 29 shows a variation of how a therapeutic tool may be utilized with an expandable imaging balloon.

FIGS. 30A and 30B show alternative configurations for therapeutic instruments which may be utilized; one variation is shown having an angled instrument arm and another variation is shown with an off-axis instrument arm.

FIGS. 31A to 31C show side and end views, respectively, of an imaging system which may be utilized with an ablation probe.

FIGS. 32A and 32B show side and end views, respectively, of another variation of the imaging hood with an ablation probe, where the imaging hood may be enclosed for regulating a temperature of the underlying tissue.

FIGS. 33A and 33B show an example in which the imaging fluid itself may be altered in temperature to facilitate various procedures upon the underlying tissue.

FIGS. 34A and 34B show an example of a laser ring generator which may be utilized with the imaging system and an example for applying the laser ring generator within the left atrium of a heart for treating atrial fibrillation.

FIGS. 35A to 35C show an example of an extendible cannula generally comprising an elongate tubular member which may be positioned within the deployment catheter during delivery and then projected distally through the imaging hood and optionally beyond.

FIGS. 36A and 36B show side and end views, respectively, of an imaging hood having one or more tubular support members integrated with the hood for passing instruments or tools therethrough for treatment upon the underlying tissue.

FIGS. 37A and 37B illustrate how an imaging device may be guided within a heart chamber to a region of interest utilizing a lighted probe positioned temporarily within, e.g., a lumen of the coronary sinus.

FIGS. 38A and 38B show an imaging hood having a removable disk-shaped member for implantation upon the tissue surface.

FIGS. 39A to 39C show one method for implanting the removable disk of FIGS. 38A and 38B.

FIGS. 40A and 40B illustrate an imaging hood having a deployable anchor assembly attached to the tissue contact edge and an assembly view of the anchors and the suture or wire connected to the anchors, respectively.

FIGS. 41A to 41D show one method for deploying the anchor assembly of FIGS. 40A and 40B for closing an opening or wound.

FIG. 42 shows another variation in which the imaging system may be fluidly coupled to a dialysis unit for filtering a patient's blood.

FIGS. 43A and 43B show a variation of the deployment catheter having a first deployable hood and a second deployable hood positioned distal to the first hood; the deployment catheter may also have a side-viewing imaging element positioned between the first and second hoods for imaging tissue between the expanded hoods.

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FIGS. 44A and 44B show side and end views, respectively, of a deployment catheter having a side-imaging balloon in an un-inflated low-profile configuration.

FIGS. 45A to 45C show side, top, and end views, respectively, of the inflated balloon of FIGS. 44A and 44B defining a visualization field in the inflated balloon.

FIGS. 46A and 46B show side and cross-sectional end views, respectively, for one method of use in visualizing a lesion upon a vessel wall within the visualization field of the inflated balloon from FIGS. 45A to 45C.

DETAILED DESCRIPTION OF THE INVENTION

A tissue-imaging and manipulation apparatus described below is able to provide real-time images in vivo of tissue regions within a body lumen such as a heart, which is filled with blood flowing dynamically therethrough and is also able to provide intravascular tools and instruments for performing various procedures upon the imaged tissue regions. Such an apparatus may be utilized for many procedures, e.g., facilitating trans-septal access to the left atrium, cannulating the coronary sinus, diagnosis of valve regurgitation/stenosis, valvuloplasty, atrial appendage closure, arrhythmogenic focus ablation, among other procedures.

One variation of a tissue access and imaging apparatus is shown in the detail perspective views of FIGS. 1A to 1C. As shown in FIG. 1A, tissue imaging and manipulation assembly 10 may be delivered intravascularly through the patient's body in a low-profile configuration via a delivery catheter or sheath 14. In the case of treating tissue, such as the mitral valve located at the outflow tract of the left atrium of the heart, it is generally desirable to enter or access the left atrium while minimizing trauma to the patient. To non-operatively effect such access, one conventional approach involves puncturing the intra-atrial septum from the right atrial chamber to the left atrial chamber in a procedure commonly called a trans-septal procedure or septostomy. For procedures such as percutaneous valve repair and replacement, trans-septal access to the left atrial chamber of the heart may allow for larger devices to be introduced into the venous system than can generally be introduced percutaneously into the arterial system.

When the imaging and manipulation assembly 10 is ready to be utilized for imaging tissue, imaging hood 12 may be advanced relative to catheter 14 and deployed from a distal opening of catheter 14, as shown by the arrow. Upon deployment, imaging hood 12 may be unconstrained to expand or open into a deployed imaging configuration, as shown in FIG. 1B. Imaging hood 12 may be fabricated from a variety of pliable or conformable biocompatible material including but not limited to, e.g., polymeric, plastic, or woven materials. One example of a woven material is Kevlar® (E. I. du Pont de Nemours, Wilmington, Del.), which is an aramid and which can be made into thin, e.g., less than 0.001 in., materials which maintain enough integrity for such applications described herein. Moreover, the imaging hood 12 may be fabricated from a translucent or opaque material and in a variety of different colors to optimize or attenuate any reflected lighting from surrounding fluids or structures, i.e., anatomical or mechanical structures or instruments. In either case, imaging hood 12 may be fabricated into a uniform structure or a scaffold-supported structure, in which case a scaffold made of a shape memory

alloy, such as Nitinol, or a spring steel, or plastic, etc., may be fabricated and covered with the polymeric, plastic, or woven material.

Imaging hood 12 may be attached at interface 24 to a deployment catheter 16 which may be translated independently of deployment catheter or sheath 14. Attachment of interface 24 may be accomplished through any number of conventional methods. Deployment catheter 16 may define a fluid delivery lumen 18 as well as an imaging lumen 20 within which an optical imaging fiber or assembly may be disposed for imaging tissue. When deployed, imaging hood 12 may expand into any number of shapes, e.g., cylindrical, conical as shown, semi-spherical, etc., provided that an open area or field 26 is defined by imaging hood 12. The open area 26 is the area within which the tissue region of interest may be imaged. Imaging hood 12 may also define an atraumatic contact lip or edge 22 for placement or abutment against the tissue region of interest. Moreover, the diameter of imaging hood 12 at its maximum fully deployed diameter, e.g., at contact lip or edge 22, is typically greater relative to a diameter of the deployment catheter 16 (although a diameter of contact lip or edge 22 may be made to have a smaller or equal diameter of deployment catheter 16). For instance, the contact edge diameter may range anywhere from 1 to 5 times (or even greater, as practicable) a diameter of deployment catheter 16. FIG. 1C shows an end view of the imaging hood 12 in its deployed configuration. Also shown are the contact lip or edge 22 and fluid delivery lumen 18 and imaging lumen 20.

The imaging and manipulation assembly 10 may additionally define a guidewire lumen therethrough, e.g., a concentric or eccentric lumen, as shown in the side and end views, respectively, of FIGS. 1D to 1F. The deployment catheter 16 may define guidewire lumen 19 for facilitating the passage of the system over or along a guidewire 17, which may be advanced intravascularly within a body lumen. The deployment catheter 16 may then be advanced over the guidewire 17, as generally known in the art.

During operation, after imaging hood 12 has been deployed, as in FIG. 1B, and desirably positioned against the tissue region to be imaged along contact edge 22, the displacing fluid may be pumped at positive pressure through fluid delivery lumen 18 until the fluid fills open area 26 completely and displaces any fluid 28 from within open area 26. The displacing fluid flow may be laminarized to improve its clearing effect and to help prevent blood from re-entering the imaging hood 12. Alternatively, fluid flow may be started before the deployment takes place. The displacing fluid, also described herein as imaging fluid, may comprise any bio-compatible fluid, e.g., saline, water, plasma, etc., which is sufficiently transparent to allow for relatively undistorted visualization through the fluid. Alternatively or additionally, any number of therapeutic drugs may be suspended within the fluid or may comprise the fluid itself which is pumped into open area 26 and which is subsequently passed into and through the heart and the patient body.

As seen in the example of FIGS. 2A and 2B, deployment catheter 16 may be manipulated to position deployed imaging hood 12 against or near the underlying tissue region of interest to be imaged, in this example a portion of annulus A of mitral valve MV within the left atrial chamber. As the surrounding blood 30 flows around imaging hood 12 and within open area 26 defined within imaging hood 12, as seen in FIG. 2A, the underlying annulus A is obstructed by the opaque blood 30 and is difficult to view through the imaging lumen 20. The translucent fluid 28, such as saline, may then be pumped through fluid delivery lumen 18, intermittently

or continuously, until the blood 30 is at least partially, and preferably completely, displaced from within open area 26 by fluid 28, as shown in FIG. 2B.

Although contact edge 22 need not directly contact the underlying tissue, it is at least preferably brought into close proximity to the tissue such that the flow of clear fluid 28 from open area 26 may be maintained to inhibit significant backflow of blood 30 back into open area 26. Contact edge 22 may also be made of a soft elastomeric material such as certain soft grades of silicone or polyurethane, as typically known, to help contact edge 22 conform to an uneven or rough underlying anatomical tissue surface. Once the blood 30 has been displaced from imaging hood 12, an image may then be viewed of the underlying tissue through the clear fluid 30. This image may then be recorded or available for real-time viewing for performing a therapeutic procedure. The positive flow of fluid 28 may be maintained continuously to provide for clear viewing of the underlying tissue. Alternatively, the fluid 28 may be pumped temporarily or sporadically only until a clear view of the tissue is available to be imaged and recorded, at which point the fluid flow 28 may cease and blood 30 may be allowed to seep or flow back into imaging hood 12. This process may be repeated a number of times at the same tissue region or at multiple tissue regions.

In desirably positioning the assembly at various regions within the patient body, a number of articulation and manipulation controls may be utilized. For example, as shown in the articulatable imaging assembly 40 in FIG. 3A, one or more push-pull wires 42 may be routed through deployment catheter 16 for steering the distal end portion of the device in various directions 46 to desirably position the imaging hood 12 adjacent to a region of tissue to be visualized. Depending upon the positioning and the number of push-pull wires 42 utilized, deployment catheter 16 and imaging hood 12 may be articulated into any number of configurations 44. The push-pull wire or wires 42 may be articulated via their proximal ends from outside the patient body manually utilizing one or more controls. Alternatively, deployment catheter 16 may be articulated by computer control, as further described below.

Additionally or alternatively, an articulatable delivery catheter 48, which may be articulated via one or more push-pull wires and having an imaging lumen and one or more working lumens, may be delivered through the deployment catheter 16 and into imaging hood 12. With a distal portion of articulatable delivery catheter 48 within imaging hood 12, the clear displacing fluid may be pumped through delivery catheter 48 or deployment catheter 16 to clear the field within imaging hood 12. As shown in FIG. 3B, the articulatable delivery catheter 48 may be articulated within the imaging hood to obtain a better image of tissue adjacent to the imaging hood 12. Moreover, articulatable delivery catheter 48 may be articulated to direct an instrument or tool passed through the catheter 48, as described in detail below, to specific areas of tissue imaged through imaging hood 12 without having to reposition deployment catheter 16 and re-clear the imaging field within hood 12.

Alternatively, rather than passing an articulatable delivery catheter 48 through the deployment catheter 16, a distal portion of the deployment catheter 16 itself may comprise a distal end 49 which is articulatable within imaging hood 12, as shown in FIG. 3C. Directed imaging, instrument delivery, etc., may be accomplished directly through one or more lumens within deployment catheter 16 to specific regions of the underlying tissue imaged within imaging hood 12.

Visualization within the imaging hood **12** may be accomplished through an imaging lumen **20** defined through deployment catheter **16**, as described above. In such a configuration, visualization is available in a straight-line manner, i.e., images are generated from the field distally along a longitudinal axis defined by the deployment catheter **16**. Alternatively or additionally, an articulatable imaging assembly having a pivotable support member **50** may be connected to, mounted to, or otherwise passed through deployment catheter **16** to provide for visualization off-axis relative to the longitudinal axis defined by deployment catheter **16**, as shown in FIG. 4A. Support member **50** may have an imaging element **52**, e.g., a CCD or CMOS imager or optical fiber, attached at its distal end with its proximal end connected to deployment catheter **16** via a pivoting connection **54**.

If one or more optical fibers are utilized for imaging, the optical fibers **58** may be passed through deployment catheter **16**, as shown in the cross-section of FIG. 4B, and routed through the support member **50**. The use of optical fibers **58** may provide for increased diameter sizes of the one or several lumens **56** through deployment catheter **16** for the passage of diagnostic and/or therapeutic tools therethrough. Alternatively, electronic chips, such as a charge coupled device (CCD) or a CMOS imager, which are typically known, may be utilized in place of the optical fibers **58**, in which case the electronic imager may be positioned in the distal portion of the deployment catheter **16** with electric wires being routed proximally through the deployment catheter **16**. Alternatively, the electronic imagers may be wirelessly coupled to a receiver for the wireless transmission of images. Additional optical fibers or light emitting diodes (LEDs) can be used to provide lighting for the image or operative theater, as described below in further detail. Support member **50** may be pivoted via connection **54** such that the member **50** can be positioned in a low-profile configuration within channel or groove **60** defined in a distal portion of catheter **16**, as shown in the cross-section of FIG. 4C. During intravascular delivery of deployment catheter **16** through the patient body, support member **50** can be positioned within channel or groove **60** with imaging hood **12** also in its low-profile configuration. During visualization, imaging hood **12** may be expanded into its deployed configuration and support member **50** may be deployed into its off-axis configuration for imaging the tissue adjacent to hood **12**, as in FIG. 4A. Other configurations for support member **50** for off-axis visualization may be utilized, as desired.

FIG. 5 shows an illustrative cross-sectional view of a heart H having tissue regions of interest being viewed via an imaging assembly **10**. In this example, delivery catheter assembly **70** may be introduced percutaneously into the patient's vasculature and advanced through the superior vena cava SVC and into the right atrium RA. The delivery catheter or sheath **72** may be articulated through the atrial septum AS and into the left atrium LA for viewing or treating the tissue, e.g., the annulus A, surrounding the mitral valve MV. As shown, deployment catheter **16** and imaging hood **12** may be advanced out of delivery catheter **72** and brought into contact or in proximity to the tissue region of interest. In other examples, delivery catheter assembly **70** may be advanced through the inferior vena cava IVC, if so desired. Moreover, other regions of the heart H, e.g., the right ventricle RV or left ventricle LV, may also be accessed and imaged or treated by imaging assembly **10**.

In accessing regions of the heart H or other parts of the body, the delivery catheter or sheath **14** may comprise a

conventional intra-vascular catheter or an endoluminal delivery device. Alternatively, robotically-controlled delivery catheters may also be optionally utilized with the imaging assembly described herein, in which case a computer-controller **74** may be used to control the articulation and positioning of the delivery catheter **14**. An example of a robotically-controlled delivery catheter which may be utilized is described in further detail in U.S. Pat. Pub. 2002/0087169 A1 to Brock et al. entitled "Flexible Instrument", which is incorporated herein by reference in its entirety. Other robotically-controlled delivery catheters manufactured by Hansen Medical, Inc. (Mountain View, Calif.) may also be utilized with the delivery catheter **14**.

To facilitate stabilization of the deployment catheter **16** during a procedure, one or more inflatable balloons or anchors **76** may be positioned along the length of catheter **16**, as shown in FIG. 6A. For example, when utilizing a trans-septal approach across the atrial septum AS into the left atrium LA, the inflatable balloons **76** may be inflated from a low-profile into their expanded configuration to temporarily anchor or stabilize the catheter **16** position relative to the heart H. FIG. 6B shows a first balloon **78** inflated while FIG. 6C also shows a second balloon **80** inflated proximal to the first balloon **78**. In such a configuration, the septal wall AS may be wedged or sandwiched between the balloons **78**, **80** to temporarily stabilize the catheter **16** and imaging hood **12**. A single balloon **78** or both balloons **78**, **80** may be used. Other alternatives may utilize expandable mesh members, malecots, or any other temporary expandable structure. After a procedure has been accomplished, the balloon assembly **76** may be deflated or re-configured into a low-profile for removal of the deployment catheter **16**.

To further stabilize a position of the imaging hood **12** relative to a tissue surface to be imaged, various anchoring mechanisms may be optionally employed for temporarily holding the imaging hood **12** against the tissue. Such anchoring mechanisms may be particularly useful for imaging tissue which is subject to movement, e.g., when imaging tissue within the chambers of a beating heart. A tool delivery catheter **82** having at least one instrument lumen and an optional visualization lumen may be delivered through deployment catheter **16** and into an expanded imaging hood **12**. As the imaging hood **12** is brought into contact against a tissue surface T to be examined, an anchoring mechanism such as a helical tissue piercing device **84** may be passed through the tool delivery catheter **82**, as shown in FIG. 7A, and into imaging hood **12**.

The helical tissue engaging device **84** may be torqued from its proximal end outside the patient body to temporarily anchor itself into the underlying tissue surface T. Once embedded within the tissue T, the helical tissue engaging device **84** may be pulled proximally relative to deployment catheter **16** while the deployment catheter **16** and imaging hood **12** are pushed distally, as indicated by the arrows in FIG. 7B, to gently force the contact edge or lip **22** of imaging hood against the tissue T. The positioning of the tissue engaging device **84** may be locked temporarily relative to the deployment catheter **16** to ensure secure positioning of the imaging hood **12** during a diagnostic or therapeutic procedure within the imaging hood **12**. After a procedure, tissue engaging device **84** may be disengaged from the tissue by torquing its proximal end in the opposite direction to remove the anchor from the tissue T and the deployment catheter **16** may be repositioned to another region of tissue where the anchoring process may be repeated or removed from the patient body. The tissue engaging device **84** may

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also be constructed from other known tissue engaging devices such as vacuum-assisted engagement or grasper-assisted engagement tools, among others.

Although a helical anchor **84** is shown, this is intended to be illustrative and other types of temporary anchors may be utilized, e.g., hooked or barbed anchors, graspers, etc. Moreover, the tool delivery catheter **82** may be omitted entirely and the anchoring device may be delivered directly through a lumen defined through the deployment catheter **16**.

In another variation where the tool delivery catheter **82** may be omitted entirely to temporarily anchor imaging hood **12**, FIG. 7C shows an imaging hood **12** having one or more tubular support members **86**, e.g. four support members **86** as shown, integrated with the imaging hood **12**. The tubular support members **86** may define lumens therethrough each having helical tissue engaging devices **88** positioned within. When an expanded imaging hood **12** is to be temporarily anchored to the tissue, the helical tissue engaging devices **88** may be urged distally to extend from imaging hood **12** and each may be torqued from its proximal end to engage the underlying tissue **T**. Each of the helical tissue engaging devices **88** may be advanced through the length of deployment catheter **16** or they may be positioned within tubular support members **86** during the delivery and deployment of imaging hood **12**. Once the procedure within imaging hood **12** is finished, each of the tissue engaging devices **88** may be disengaged from the tissue and the imaging hood **12** may be repositioned to another region of tissue or removed from the patient body.

An illustrative example is shown in FIG. 8A of a tissue imaging assembly connected to a fluid delivery system **90** and to an optional processor **98** and image recorder and/or viewer **100**. The fluid delivery system **90** may generally comprise a pump **92** and an optional valve **94** for controlling the flow rate of the fluid into the system. A fluid reservoir **96**, fluidly connected to pump **92**, may hold the fluid to be pumped through imaging hood **12**. An optional central processing unit or processor **98** may be in electrical communication with fluid delivery system **90** for controlling flow parameters such as the flow rate and/or velocity of the pumped fluid. The processor **98** may also be in electrical communication with an image recorder and/or viewer **100** for directly viewing the images of tissue received from within imaging hood **12**. Imager recorder and/or viewer **100** may also be used not only to record the image but also the location of the viewed tissue region, if so desired.

Optionally, processor **98** may also be utilized to coordinate the fluid flow and the image capture. For instance, processor **98** may be programmed to provide for fluid flow from reservoir **96** until the tissue area has been displaced of blood to obtain a clear image. Once the image has been determined to be sufficiently clear, either visually by a practitioner or by computer, an image of the tissue may be captured automatically by recorder **100** and pump **92** may be automatically stopped or slowed by processor **98** to cease the fluid flow into the patient. Other variations for fluid delivery and image capture are, of course, possible and the aforementioned configuration is intended only to be illustrative and not limiting.

FIG. 8B shows a further illustration of a hand-held variation of the fluid delivery and tissue manipulation system **110**. In this variation, system **110** may have a housing or handle assembly **112** which can be held or manipulated by the physician from outside the patient body. The fluid reservoir **114**, shown in this variation as a syringe, can be fluidly coupled to the handle assembly **112** and actuated via a pumping mechanism **116**, e.g., lead screw. Fluid reservoir

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114 may be a simple reservoir separated from the handle assembly **112** and fluidly coupled to handle assembly **112** via one or more tubes. The fluid flow rate and other mechanisms may be metered by the electronic controller **118**.

Deployment of imaging hood **12** may be actuated by a hood deployment switch **120** located on the handle assembly **112** while dispensation of the fluid from reservoir **114** may be actuated by a fluid deployment switch **122**, which can be electrically coupled to the controller **118**. Controller **118** may also be electrically coupled to a wired or wireless antenna **124** optionally integrated with the handle assembly **112**, as shown in the figure. The wireless antenna **124** can be used to wirelessly transmit images captured from the imaging hood **12** to a receiver, e.g., via Bluetooth® wireless technology (Bluetooth SIG. Inc., Bellevue, Wash.), RF, etc., for viewing on a monitor **128** or for recording for later viewing.

Articulation control of the deployment catheter **16**, or a delivery catheter or sheath **14** through which the deployment catheter **16** may be delivered, may be accomplished by computer control, as described above, in which case an additional controller may be utilized with handle assembly **112**. In the case of manual articulation, handle assembly **112** may incorporate one or more articulation controls **126** for manual manipulation of the position of deployment catheter **16**. Handle assembly **112** may also define one or more instrument ports **130** through which a number of intravascular tools may be passed for tissue manipulation and treatment within imaging hood **12**, as described further below. Furthermore, in certain procedures, fluid or debris may be sucked into imaging hood **12** for evacuation from the patient body by optionally fluidly coupling a suction pump **132** to handle assembly **112** or directly to deployment catheter **16**.

As described above, fluid may be pumped continuously into imaging hood **12** to provide for clear viewing of the underlying tissue. Alternatively, fluid may be pumped temporarily or sporadically only until a clear view of the tissue is available to be imaged and recorded, at which point the fluid flow may cease and the blood may be allowed to seep or flow back into imaging hood **12**. FIGS. 9A to 9C illustrate an example of capturing several images of the tissue at multiple regions. Deployment catheter **16** may be desirably positioned and imaging hood **12** deployed and brought into position against a region of tissue to be imaged, in this example the tissue surrounding a mitral valve MV within the left atrium of a patient's heart. The imaging hood **12** may be optionally anchored to the tissue, as described above, and then cleared by pumping the imaging fluid into the hood **12**. Once sufficiently clear, the tissue may be visualized and the image captured by control electronics **118**. The first captured image **140** may be stored and/or transmitted wirelessly **124** to a monitor **128** for viewing by the physician, as shown in FIG. 9A.

The deployment catheter **16** may be then repositioned to an adjacent portion of mitral valve MV, as shown in FIG. 9B, where the process may be repeated to capture a second image **142** for viewing and/or recording. The deployment catheter **16** may again be repositioned to another region of tissue, as shown in FIG. 9C, where a third image **144** may be captured for viewing and/or recording. This procedure may be repeated as many times as necessary for capturing a comprehensive image of the tissue surrounding mitral valve MV, or any other tissue region. When the deployment catheter **16** and imaging hood **12** is repositioned from tissue region to tissue region, the pump may be stopped during

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positioning and blood or surrounding fluid may be allowed to enter within imaging hood 12 until the tissue is to be imaged, where the imaging hood 12 may be cleared, as above.

As mentioned above, when the imaging hood 12 is cleared by pumping the imaging fluid within for clearing the blood or other bodily fluid, the fluid may be pumped continuously to maintain the imaging fluid within the hood 12 at a positive pressure or it may be pumped under computer control for slowing or stopping the fluid flow into the hood 12 upon detection of various parameters or until a clear image of the underlying tissue is obtained. The control electronics 118 may also be programmed to coordinate the fluid flow into the imaging hood 12 with various physical parameters to maintain a clear image within imaging hood 12.

One example is shown in FIG. 10A which shows a chart 150 illustrating how fluid pressure within the imaging hood 12 may be coordinated with the surrounding blood pressure. Chart 150 shows the cyclical blood pressure 156 alternating between diastolic pressure 152 and systolic pressure 154 over time T due to the beating motion of the patient heart. The fluid pressure of the imaging fluid, indicated by plot 160, within imaging hood 12 may be automatically timed to correspond to the blood pressure changes 160 such that an increased pressure is maintained within imaging hood 12 which is consistently above the blood pressure 156 by a slight increase ΔP , as illustrated by the pressure difference at the peak systolic pressure 158. This pressure difference, ΔP , may be maintained within imaging hood 12 over the pressure variance of the surrounding blood pressure to maintain a positive imaging fluid pressure within imaging hood 12 to maintain a clear view of the underlying tissue. One benefit of maintaining a constant ΔP is a constant flow and maintenance of a clear field.

FIG. 10B shows a chart 162 illustrating another variation for maintaining a clear view of the underlying tissue where one or more sensors within the imaging hood 12, as described in further detail below, may be configured to sense pressure changes within the imaging hood 12 and to correspondingly increase the imaging fluid pressure within imaging hood 12. This may result in a time delay, ΔT , as illustrated by the shifted fluid pressure 160 relative to the cycling blood pressure 156, although the time delay ΔT may be negligible in maintaining the clear image of the underlying tissue. Predictive software algorithms can also be used to substantially eliminate this time delay by predicting when the next pressure wave peak will arrive and by increasing the pressure ahead of the pressure wave's arrival by an amount of time equal to the aforementioned time delay to essentially cancel the time delay out.

The variations in fluid pressure within imaging hood 12 may be accomplished in part due to the nature of imaging hood 12. An inflatable balloon, which is conventionally utilized for imaging tissue, may be affected by the surrounding blood pressure changes. On the other hand, an imaging hood 12 retains a constant volume therewithin and is structurally unaffected by the surrounding blood pressure changes, thus allowing for pressure increases therewithin. The material that hood 12 is made from may also contribute to the manner in which the pressure is modulated within this hood 12. A stiffer hood material, such as high durometer polyurethane or Nylon, may facilitate the maintaining of an open hood when deployed. On the other hand, a relatively lower durometer or softer material, such as a low durometer PVC or polyurethane, may collapse from the surrounding fluid pressure and may not adequately maintain a deployed or expanded hood.

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Turning now to the imaging hood, other variations of the tissue imaging assembly may be utilized, as shown in FIG. 11A, which shows another variation comprising an additional imaging balloon 172 within an imaging hood 174. In this variation, an expandable balloon 172 having a translucent skin may be positioned within imaging hood 174. Balloon 172 may be made from any distensible biocompatible material having sufficient translucent properties which allow for visualization therethrough. Once the imaging hood 174 has been deployed against the tissue region of interest, balloon 172 may be filled with a fluid, such as saline, or less preferably a gas, until balloon 172 has been expanded until the blood has been sufficiently displaced. The balloon 172 may this be expanded proximal to or into contact against the tissue region to be viewed. The balloon 172 can also be filled with contrast media to allow it to be viewed on fluoroscopy to aid in its positioning. The imager, e.g., fiber optic, positioned within deployment catheter 170 may then be utilized to view the tissue region through the balloon 172 and any additional fluid which may be pumped into imaging hood 174 via one or more optional fluid ports 176, which may be positioned proximally of balloon 172 along a portion of deployment catheter 170. Alternatively, balloon 172 may define one or more holes over its surface which allow for seepage or passage of the fluid contained therein to escape and displace the blood from within imaging hood 174.

FIG. 11B shows another alternative in which balloon 180 may be utilized alone. Balloon 180, attached to deployment catheter 178, may be filled with fluid, such as saline or contrast media, and is preferably allowed to come into direct contact with the tissue region to be imaged.

FIG. 12A shows another alternative in which deployment catheter 16 incorporates imaging hood 12, as above, and includes an additional flexible membrane 182 within imaging hood 12. Flexible membrane 182 may be attached at a distal end of catheter 16 and optionally at contact edge 22. Imaging hood 12 may be utilized, as above, and membrane 182 may be deployed from catheter 16 in vivo or prior to placing catheter 16 within a patient to reduce the volume within imaging hood 12. The volume may be reduced or minimized to reduce the amount of fluid dispensed for visualization or simply be to reduced depending upon the area of tissue to be visualized.

FIGS. 12B and 12C show yet another alternative in which imaging hood 186 may be withdrawn proximally within deployment catheter 184 or deployed distally from catheter 186, as shown, to vary the volume of imaging hood 186 and thus the volume of dispensed fluid. Imaging hood 186 may be seen in FIG. 12B as being partially deployed from, e.g., a circumferentially defined lumen within catheter 184, such as annular lumen 188. The underlying tissue may be visualized with imaging hood 186 only partially deployed. Alternatively, imaging hood 186' may be fully deployed, as shown in FIG. 12C, by urging hood 186' distally out from annular lumen 188. In this expanded configuration, the area of tissue to be visualized may be increased as hood 186' is expanded circumferentially.

FIGS. 13A and 13B show perspective and cross-sectional side views, respectively, of yet another variation of imaging assembly which may utilize a fluid suction system for minimizing the amount of fluid injected into the patient's heart or other body lumen during tissue visualization. Deployment catheter 190 in this variation may define an inner tubular member 196 which may be integrated with deployment catheter 190 or independently translatable. Fluid delivery lumen 198 defined through member 196 may be fluidly connected to imaging hood 192, which may also

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define one or more open channels **194** over its contact lip region. Fluid pumped through fluid delivery lumen **198** may thus fill open area **202** to displace any blood or other fluids or objects therewithin. As the clear fluid is forced out of open area **202**, it may be sucked or drawn immediately through one or more channels **194** and back into deployment catheter **190**. Tubular member **196** may also define one or more additional working channels **200** for the passage of any tools or visualization devices.

In deploying the imaging hood in the examples described herein, the imaging hood may take on any number of configurations when positioned or configured for a low-profile delivery within the delivery catheter, as shown in the examples of FIGS. **14A** to **14D**. These examples are intended to be illustrative and are not intended to be limiting in scope. FIG. **14A** shows one example in which imaging hood **212** may be compressed within catheter **210** by folding hood **212** along a plurality of pleats. Hood **212** may also comprise scaffolding or frame **214** made of a super-elastic or shape memory material or alloy, e.g., Nitinol, Elgiloy, shape memory polymers, electroactive polymers, or a spring stainless steel. The shape memory material may act to expand or deploy imaging hood **212** into its expanded configuration when urged in the direction of the arrow from the constraints of catheter **210**.

FIG. **14B** shows another example in which imaging hood **216** may be expanded or deployed from catheter **210** from a folded and overlapping configuration. Frame or scaffolding **214** may also be utilized in this example. FIG. **14C** shows yet another example in which imaging hood **218** may be rolled, inverted, or everted upon itself for deployment. In yet another example, FIG. **14D** shows a configuration in which imaging hood **220** may be fabricated from an extremely compliant material which allows for hood **220** to be simply compressed into a low-profile shape. From this low-profile compressed shape, simply releasing hood **220** may allow for it to expand into its deployed configuration, especially if a scaffold or frame of a shape memory or superelastic material, e.g., Nitinol, is utilized in its construction.

Another variation for expanding the imaging hood is shown in FIGS. **15A** and **15B** which illustrates an helically expanding frame or support **230**. In its constrained low-profile configuration, shown in FIG. **15A**, helical frame **230** may be integrated with the imaging hood **12** membrane. When free to expand, as shown in FIG. **15B**, helical frame **230** may expand into a conical or tapered shape. Helical frame **230** may alternatively be made out of heat-activated Nitinol to allow it to expand upon application of a current.

FIGS. **16A** and **16B** show yet another variation in which imaging hood **12** may comprise one or more hood support members **232** integrated with the hood membrane. These longitudinally attached support members **232** may be pivotably attached at their proximal ends to deployment catheter **16**. One or more pullwires **234** may be routed through the length of deployment catheter **16** and extend through one or more openings **238** defined in deployment catheter **16** proximally to imaging hood **12** into attachment with a corresponding support member **232** at a pullwire attachment point **236**. The support members **232** may be fabricated from a plastic or metal, such as stainless steel. Alternatively, the support members **232** may be made from a superelastic or shape memory alloy, such as Nitinol, which may self-expand into its deployed configuration without the use or need of pullwires. A heat-activated Nitinol may also be used which expands upon the application of thermal energy or electrical energy. In another alternative, support members **232** may

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also be constructed as inflatable lumens utilizing, e.g., PET balloons. From its low-profile delivery configuration shown in FIG. **16A**, the one or more pullwires **234** may be tensioned from their proximal ends outside the patient body to pull a corresponding support member **232** into a deployed configuration, as shown in FIG. **16B**, to expand imaging hood **12**. To reconfigure imaging hood **12** back into its low profile, deployment catheter **16** may be pulled proximally into a constraining catheter or the pullwires **234** may be simply pushed distally to collapse imaging hood **12**.

FIGS. **17A** and **17B** show yet another variation of imaging hood **240** having at least two or more longitudinally positioned support members **242** supporting the imaging hood membrane. The support members **242** each have cross-support members **244** which extend diagonally between and are pivotably attached to the support members **242**. Each of the cross-support members **244** may be pivotably attached to one another where they intersect between the support members **242**. A jack or screw member **246** may be coupled to each cross-support member **244** at this intersection point and a torquing member, such as a torqueable wire **248**, may be coupled to each jack or screw member **246** and extend proximally through deployment catheter **16** to outside the patient body. From outside the patient body, the torqueable wires **248** may be torqued to turn the jack or screw member **246** which in turn urges the cross-support members **244** to angle relative to one another and thereby urge the support members **242** away from one another. Thus, the imaging hood **240** may be transitioned from its low-profile, shown in FIG. **17A**, to its expanded profile, shown in FIG. **17B**, and back into its low-profile by torquing wires **248**.

FIGS. **18A** and **18B** show yet another variation on the imaging hood and its deployment. As shown, a distal portion of deployment catheter **16** may have several pivoting members **250**, e.g., two to four sections, which form a tubular shape in its low profile configuration, as shown in FIG. **18A**. When pivoted radially about deployment catheter **16**, pivoting members **250** may open into a deployed configuration having distensible or expanding membranes **252** extending over the gaps in-between the pivoting members **250**, as shown in FIG. **18B**. The distensible membrane **252** may be attached to the pivoting members **250** through various methods, e.g., adhesives, such that when the pivoting members **250** are fully extended into a conical shape, the pivoting members **250** and membrane **252** form a conical shape for use as an imaging hood. The distensible membrane **252** may be made out of a porous material such as a mesh or PTFE or out of a translucent or transparent polymer such as polyurethane, PVC, Nylon, etc.

FIGS. **19A** and **19B** show yet another variation where the distal portion of deployment catheter **16** may be fabricated from a flexible metallic or polymeric material to form a radially expanding hood **254**. A plurality of slots **256** may be formed in a uniform pattern over the distal portion of deployment catheter **16**, as shown in FIG. **19A**. The slots **256** may be formed in a pattern such that when the distal portion is urged radially open, utilizing any of the methods described above, a radially expanded and conically-shaped hood **254** may be formed by each of the slots **256** expanding into an opening, as shown in FIG. **19B**. A distensible membrane **258** may overlie the exterior surface or the interior surface of the hood **254** to form a fluid-impermeable hood **254** such that the hood **254** may be utilized as an imaging hood. Alternatively, the distensible membrane **258** may alternatively be formed in each opening **258** to form the

fluid-impermeable hood **254**. Once the imaging procedure has been completed, hood **254** may be retracted into its low-profile configuration.

Yet another configuration for the imaging hood may be seen in FIGS. **20A** and **20B** where the imaging hood may be formed from a plurality of overlapping hood members **260** which overlies one another in an overlapping pattern. When expanded, each of the hood members **260** may extend radially outward relative to deployment catheter **16** to form a conically-shaped imaging hood, as shown in FIG. **20B**. Adjacent hood members **260** may overlap one another along an overlapping interface **262** to form a fluid-retaining surface within the imaging hood. Moreover, the hood members **260** may be made from any number of biocompatible materials, e.g., Nitinol, stainless steel, polymers, etc., which are sufficiently strong to optionally retract surrounding tissue from the tissue region of interest.

Although it is generally desirable to have an imaging hood contact against a tissue surface in a normal orientation, the imaging hood may be alternatively configured to contact the tissue surface at an acute angle. An imaging hood configured for such contact against tissue may also be especially suitable for contact against tissue surfaces having an unpredictable or uneven anatomical geography. For instance, as shown in the variation of FIG. **21A**, deployment catheter **270** may have an imaging hood **272** that is configured to be especially compliant. In this variation, imaging hood **272** may be comprised of one or more sections **274** that are configured to fold or collapse, e.g. by utilizing a pleated surface. Thus, as shown in FIG. **21B**, when imaging hood **272** is contacted against uneven tissue surface **T**, sections **274** are able to conform closely against the tissue. These sections **274** may be individually collapsible by utilizing an accordion style construction to allow conformation, e.g. to the trabeculae in the heart or the uneven anatomy that may be found inside the various body lumens.

In yet another alternative, FIG. **22A** shows another variation in which an imaging hood **282** is attached to deployment catheter **280**. The contact lip or edge **284** may comprise one or more electrical contacts **286** positioned circumferentially around contact edge **284**. The electrical contacts **286** may be configured to contact the tissue and indicate affirmatively whether tissue contact was achieved, e.g., by measuring the differential impedance between blood and tissue. Alternatively, a processor, e.g., processor **98**, in electrical communication with contacts **286** may be configured to determine what type of tissue is in contact with electrical contacts **286**. In yet another alternative, the processor **98** may be configured to measure any electrical activity that may be occurring in the underlying tissue. e.g., accessory pathways, for the purposes of electrically mapping the cardiac tissue and subsequently treating, as described below, any arrhythmias which may be detected.

Another variation for ensuring contact between imaging hood **282** and the underlying tissue may be seen in FIG. **22B**. This variation may have an inflatable contact edge **288** around the circumference of imaging hood **282**. The inflatable contact edge **288** may be inflated with a fluid or gas through inflation lumen **289** when the imaging hood **282** is to be placed against a tissue surface having an uneven or varied anatomy. The inflated circumferential surface **288** may provide for continuous contact over the hood edge by conforming against the tissue surface and facilitating imaging fluid retention within hood **282**.

Aside from the imaging hood, various instrumentation may be utilized with the imaging and manipulation system. For instance, after the field within imaging hood **12** has been

cleared of the opaque blood and the underlying tissue is visualized through the clear fluid, blood may seep back into the imaging hood **12** and obstruct the view. One method for automatically maintaining a clear imaging field may utilize a transducer, e.g., an ultrasonic transducer **290**, positioned at the distal end of deployment catheter within the imaging hood **12**, as shown in FIG. **23**. The transducer **290** may send an energy pulse **292** into the imaging hood **12** and wait to detect back-scattered energy **294** reflected from debris or blood within the imaging hood **12**. If back-scattered energy is detected, the pump may be actuated automatically to dispense more fluid into the imaging hood until the debris or blood is no longer detected.

Alternatively, one or more sensors **300** may be positioned on the imaging hood **12** itself, as shown in FIG. **24A**, to detect a number of different parameters. For example, sensors **300** may be configured to detect for the presence of oxygen in the surrounding blood, blood and/or imaging fluid pressure, color of the fluid within the imaging hood, etc. Fluid color may be particularly useful in detecting the presence of blood within the imaging hood **12** by utilizing a reflective type sensor to detect back reflection from blood. Any reflected light from blood which may be present within imaging hood **12** may be optically or electrically transmitted through deployment catheter **16** and to a red colored filter within control electronics **118**. Any red color which may be detected may indicate the presence of blood and trigger a signal to the physician or automatically actuate the pump to dispense more fluid into the imaging hood **12** to clear the blood.

Alternative methods for detecting the presence of blood within the hood **12** may include detecting transmitted light through the imaging fluid within imaging blood **12**. If a source of white light, e.g., utilizing LEDs or optical fibers, is illuminated inside imaging hood **12**, the presence of blood may cause the color red to be filtered through this fluid. The degree or intensity of the red color detected may correspond to the amount of blood present within imaging hood **12**. A red color sensor can simply comprise, in one variation, a phototransistor with a red transmitting filter over it which can establish how much red light is detected, which in turn can indicate the presence of blood within imaging hood **12**. Once blood is detected, the system may pump more clearing fluid through and enable closed loop feedback control of the clearing fluid pressure and flow level.

Any number of sensors may be positioned along the exterior **302** of imaging hood **12** or within the interior **304** of imaging hood **12** to detect parameters not only exteriorly to imaging hood **12** but also within imaging hood **12**. Such a configuration, as shown in FIG. **24B**, may be particularly useful for automatically maintaining a clear imaging field based upon physical parameters such as blood pressure, as described above for FIGS. **10A** and **10B**.

Aside from sensors, one or more light emitting diodes (LEDs) may be utilized to provide lighting within the imaging hood **12**. Although illumination may be provided by optical fibers routed through deployment catheter **16**, the use of LEDs over the imaging hood **12** may eliminate the need for additional optical fibers for providing illumination. The electrical wires connected to the one or more LEDs may be routed through or over the hood **12** and along an exterior surface or extruded within deployment catheter **16**. One or more LEDs may be positioned in a circumferential pattern **306** around imaging hood **12**, as shown in FIG. **25A**, or in a linear longitudinal pattern **308** along imaging hood **12**, as shown in FIG. **25B**. Other patterns, such as a helical or spiral

pattern, may also be utilized. Alternatively, LEDs may be positioned along a support member forming part of imaging hood 12.

In another alternative for illumination within imaging hood 12, a separate illumination tool 310 may be utilized, as shown in FIG. 26A. An example of such a tool may comprise a flexible intravascular delivery member 312 having a carrier member 314 pivotably connected 316 to a distal end of delivery member 312. One or more LEDs 318 may be mounted along carrier member 314. In use, delivery member 312 may be advanced through deployment catheter 16 until carrier member 314 is positioned within imaging hood 12. Once within imaging hood 12, carrier member 314 may be pivoted in any number of directions to facilitate or optimize the illumination within the imaging hood 12, as shown in FIG. 26B.

In utilizing LEDs for illumination, whether positioned along imaging hood 12 or along a separate instrument, the LEDs may comprise a single LED color, e.g., white light. Alternatively, LEDs of other colors, e.g., red, blue, yellow, etc., may be utilized exclusively or in combination with white LEDs to provide for varied illumination of the tissue or fluids being imaged. Alternatively, sources of infrared or ultraviolet light may be employed to enable imaging beneath the tissue surface or cause fluorescence of tissue for use in system guidance, diagnosis, or therapy.

Aside from providing a visualization platform, the imaging assembly may also be utilized to provide a therapeutic platform for treating tissue being visualized. As shown in FIG. 27, deployment catheter 320 may have imaging hood 322, as described above, and fluid delivery lumen 324 and imaging lumen 326. In this variation, a therapeutic tool such as needle 328 may be delivered through fluid delivery lumen 324 or in another working lumen and advanced through open area 332 for treating the tissue which is visualized. In this instance, needle 328 may define one or several ports 330 for delivering drugs therethrough. Thus, once the appropriate region of tissue has been imaged and located, needle 328 may be advanced and pierced into the underlying tissue where a therapeutic agent may be delivered through ports 330. Alternatively, needle 328 may be in electrical communication with a power source 334, e.g., radio-frequency, microwave, etc., for ablating the underlying tissue area of interest.

FIG. 28 shows another alternative in which deployment catheter 340 may have imaging hood 342 attached thereto, as above, but with a therapeutic tool 344 in the configuration of a helical tissue piercing device 344. Also shown and described above in FIGS. 7A and 7B for use in stabilizing the imaging hood relative to the underlying tissue, the helical tissue piercing device 344 may also be utilized to manipulate the tissue for a variety of therapeutic procedures. The helical portion 346 may also define one or several ports for delivery of therapeutic agents therethrough.

In yet another alternative, FIG. 29 shows a deployment catheter 350 having an expandable imaging balloon 352 filled with, e.g., saline 356. A therapeutic tool 344, as above, may be translatable relative to balloon 352. To prevent the piercing portion 346 of the tool from tearing balloon 352, a stop 354 may be formed on balloon 352 to prevent the proximal passage of portion 346 past stop 354.

Alternative configurations for tools which may be delivered through deployment catheter 16 for use in tissue manipulation within imaging hood 12 are shown in FIGS. 30A and 30B. FIG. 30A shows one variation of an angled instrument 360, such as a tissue grasper, which may be configured to have an elongate shaft for intravascular deliv-

ery through deployment catheter 16 with a distal end which may be angled relative to its elongate shaft upon deployment into imaging hood 12. The elongate shaft may be configured to angle itself automatically, e.g., by the elongate shaft being made at least partially from a shape memory alloy, or upon actuation, e.g., by tensioning a pullwire. FIG. 30B shows another configuration for an instrument 362 being configured to reconfigure its distal portion into an off-axis configuration within imaging hood 12. In either case, the instruments 360, 362 may be reconfigured into a low-profile shape upon withdrawing them proximally back into deployment catheter 16.

Other instruments or tools which may be utilized with the imaging system is shown in the side and end views of FIGS. 31A to 31C. FIG. 31A shows a probe 370 having a distal end effector 372, which may be reconfigured from a low-profile shape to a curved profile. The end effector 372 may be configured as an ablation probe utilizing radio-frequency energy, microwave energy, ultrasound energy, laser energy or even cryo-ablation. Alternatively, the end effector 372 may have several electrodes upon it for detecting or mapping electrical signals transmitted through the underlying tissue.

In the case of an end effector 372 utilized for ablation of the underlying tissue, an additional temperature sensor such as a thermocouple or thermistor 374 positioned upon an elongate member 376 may be advanced into the imaging hood 12 adjacent to the distal end effector 372 for contacting and monitoring a temperature of the ablated tissue. FIG. 31B shows an example in the end view of one configuration for the distal end effector 372 which may be simply angled into a perpendicular configuration for contacting the tissue. FIG. 31C shows another example where the end effector may be reconfigured into a curved end effector 378 for increased tissue contact.

FIGS. 32A and 32B show another variation of an ablation tool utilized with an imaging hood 12 having an enclosed bottom portion. In this variation, an ablation probe, such as a cryo-ablation probe 380 having a distal end effector 382, may be positioned through the imaging hood 12 such that the end effector 382 is placed distally of a transparent membrane or enclosure 384, as shown in the end view of FIG. 32B. The shaft of probe 380 may pass through an opening 386 defined through the membrane 384. In use, the clear fluid may be pumped into imaging hood 12, as described above, and the distal end effector 382 may be placed against a tissue region to be ablated with the imaging hood 12 and the membrane 384 positioned atop or adjacent to the ablated tissue. In the case of cryo-ablation, the imaging fluid may be warmed prior to dispensing into the imaging hood 12 such that the tissue contacted by the membrane 384 may be warmed during the cryo-ablation procedure. In the case of thermal ablation, e.g., utilizing radio-frequency energy, the fluid dispensed into the imaging hood 12 may be cooled such that the tissue contacted by the membrane 384 and adjacent to the ablation probe during the ablation procedure is likewise cooled.

In either example described above, the imaging fluid may be varied in its temperature to facilitate various procedures to be performed upon the tissue. In other cases, the imaging fluid itself may be altered to facilitate various procedures. For instance as shown in FIG. 33A, a deployment catheter 16 and imaging hood 12 may be advanced within a hollow body organ, such as a bladder filled with urine 394, towards a lesion or tumor 392 on the bladder wall. The imaging hood 12 may be placed entirely over the lesion 392, or over a portion of the lesion. Once secured against the tissue wall

390, a cryo-fluid, i.e., a fluid which has been cooled to below freezing temperatures of, e.g., water or blood, may be pumped into the imaging hood 12 to cryo-ablate the lesion 390, as shown in FIG. 33B while avoiding the creation of ice on the instrument or surface of tissue.

As the cryo-fluid leaks out of the imaging hood 12 and into the organ, the fluid may be warmed naturally by the patient body and ultimately removed. The cryo-fluid may be a colorless and translucent fluid which enables visualization therethrough of the underlying tissue. An example of such a fluid is Fluorinert™ (3M, St. Paul, Minn.), which is a colorless and odorless perfluorinated liquid. The use of a liquid such as Fluorinert™ enables the cryo-ablation procedure without the formation of ice within or outside of the imaging hood 12. Alternatively, rather than utilizing cryo-ablation, hyperthermic treatments may also be effected by heating the Fluorinert™ liquid to elevated temperatures for ablating the lesion 392 within the imaging hood 12. Moreover, Fluorinert™ may be utilized in various other parts of the body, such as within the heart.

FIG. 34A shows another variation of an instrument which may be utilized with the imaging system. In this variation, a laser ring generator 400 may be passed through the deployment catheter 16 and partially into imaging hood 12. A laser ring generator 400 is typically used to create a circular ring of laser energy 402 for generating a conduction block around the pulmonary veins typically in the treatment of atrial fibrillation. The circular ring of laser energy 402 may be generated such that a diameter of the ring 402 is contained within a diameter of the imaging hood 12 to allow for tissue ablation directly upon tissue being imaged. Signals which cause atrial fibrillation typically come from the entry area of the pulmonary veins into the left atrium and treatments may sometimes include delivering ablation energy to the ostia of the pulmonary veins within the atrium. The ablated areas of the tissue may produce a circular scar which blocks the impulses for atrial fibrillation.

When using the laser energy to ablate the tissue of the heart, it may be generally desirable to maintain the integrity and health of the tissue overlying the surface while ablating the underlying tissue. This may be accomplished, for example, by cooling the imaging fluid to a temperature below the body temperature of the patient but which is above the freezing point of blood (e.g., 2.degree. C. to 35.degree. C.). The cooled imaging fluid may thus maintain the surface tissue at the cooled fluid temperature while the deeper underlying tissue remains at the patient body temperature. When the laser energy (or other types of energy such as radio frequency energy, microwave energy, ultrasound energy, etc.) irradiates the tissue, both the cooled tissue surface as well as the deeper underlying tissue will rise in temperature uniformly. The deeper underlying tissue, which was maintained at the body temperature, will increase to temperatures which are sufficiently high to destroy the underlying tissue. Meanwhile, the temperature of the cooled surface tissue will also rise but only to temperatures that are near body temperature or slightly above.

Accordingly, as shown in FIG. 34B, one example for treatment may include passing deployment catheter 16 across the atrial septum AS and into the left atrium LA of the patient's heart H. Other methods of accessing the left atrium LA may also be utilized. The imaging hood 12 and laser ring generator 400 may be positioned adjacent to or over one or more of the ostium OT of the pulmonary veins PV and the laser generator 400 may ablate the tissue around the ostium OT with the circular ring of laser energy 402 to create a conduction block. Once one or more of the tissue around the

ostium OT have been ablated, the imaging hood 12 may be reconfigured into a low profile for removal from the patient heart H.

One of the difficulties in treating tissue in or around the ostium OT is the dynamic fluid flow of blood through the ostium OT. The dynamic forces make cannulation or entry of the ostium OT difficult. Thus, another variation on instruments or tools utilizable with the imaging system is an extendible cannula 410 having a cannula lumen 412 defined therethrough, as shown in FIG. 35A. The extendible cannula 410 may generally comprise an elongate tubular member which may be positioned within the deployment catheter 16 during delivery and then projected distally through the imaging hood 12 and optionally beyond, as shown in FIG. 35B.

In use, once the imaging hood 12 has been desirably positioned relative to the tissue, e.g., as shown in FIG. 35C outside the ostium OT of a pulmonary vein PV, the extendible cannula 410 may be projected distally from the deployment catheter 16 while optionally imaging the tissue through the imaging hood 12, as described above. The extendible cannula 410 may be projected distally until its distal end is extended at least partially into the ostium OT. Once in the ostium OT, an instrument or energy ablation device may be extended through and out of the cannula lumen 412 for treatment within the ostium OT. Upon completion of the procedure, the cannula 410 may be withdrawn proximally and removed from the patient body. The extendible cannula 410 may also include an inflatable occlusion balloon at or near its distal end to block the blood flow out of the PV to maintain a clear view of the tissue region. Alternatively, the extendible cannula 410 may define a lumen therethrough beyond the occlusion balloon to bypass at least a portion of the blood that normally exits the pulmonary vein PV by directing the blood through the cannula 410 to exit proximal of the imaging hood.

Yet another variation for tool or instrument use may be seen in the side and end views of FIGS. 36A and 36B. In this variation, imaging hood 12 may have one or more tubular support members 420 integrated with the hood 12. Each of the tubular support members 420 may define an access lumen 422 through which one or more instruments or tools may be delivered for treatment upon the underlying tissue. One particular example is shown and described above for FIG. 7C.

Various methods and instruments may be utilized for using or facilitating the use of the system. For instance, one method may include facilitating the initial delivery and placement of a device into the patient's heart. In initially guiding the imaging assembly within the heart chamber to, e.g., the mitral valve MV, a separate guiding probe 430 may be utilized, as shown in FIGS. 37A and 37B. Guiding probe 430 may, for example, comprise an optical fiber through which a light source 434 may be used to illuminate a distal tip portion 432. The tip portion 432 may be advanced into the heart through, e.g., the coronary sinus CS, until the tip is positioned adjacent to the mitral valve MV. The tip 432 may be illuminated, as shown in FIG. 37A, and imaging assembly 10 may then be guided towards the illuminated tip 432, which is visible from within the atrial chamber, towards mitral valve MV.

Aside from the devices and methods described above, the imaging system may be utilized to facilitate various other procedures. Turning now to FIGS. 38A and 38B, the imaging hood of the device in particular may be utilized. In this example, a collapsible membrane or disk-shaped member 440 may be temporarily secured around the contact edge or

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lip of imaging hood 12. During intravascular delivery, the imaging hood 12 and the attached member 440 may both be in a collapsed configuration to maintain a low profile for delivery. Upon deployment, both the imaging hood 12 and the member 440 may extend into their expanded configurations.

The disk-shaped member 440 may be comprised of a variety of materials depending upon the application. For instance, member 440 may be fabricated from a porous polymeric material infused with a drug eluting medicament 442 for implantation against a tissue surface for slow infusion of the medicament into the underlying tissue. Alternatively, the member 440 may be fabricated from a non-porous material, e.g., metal or polymer, for implantation and closure of a wound or over a cavity to prevent fluid leakage. In yet another alternative, the member 440 may be made from a distensible material which is secured to imaging hood 12 in an expanded condition. Once implanted or secured on a tissue surface or wound, the expanded member 440 may be released from imaging hood 12. Upon release, the expanded member 440 may shrink to a smaller size while approximating the attached underlying tissue, e.g., to close a wound or opening.

One method for securing the disk-shaped member 440 to a tissue surface may include a plurality of tissue anchors 444, e.g., barbs, hooks, projections, etc., which are attached to a surface of the member 440. Other methods of attachments may include adhesives, suturing, etc. In use, as shown in FIGS. 39A to 39C, the imaging hood 12 may be deployed in its expanded configuration with member 440 attached thereto with the plurality of tissue anchors 444 projecting distally. The tissue anchors 444 may be urged into a tissue region to be treated 446, as seen in FIG. 39A, until the anchors 444 are secured in the tissue and member 440 is positioned directly against the tissue, as shown in FIG. 39B. A pullwire may be actuated to release the member 440 from the imaging hood 12 and deployment catheter 16 may be withdrawn proximally to leave member 440 secured against the tissue 446.

Another variation for tissue manipulation and treatment may be seen in the variation of FIG. 40A, which illustrates an imaging hood 12 having a deployable anchor assembly 450 attached to the tissue contact edge 22. FIG. 40B illustrates the anchor assembly 450 detached from the imaging hood 12 for clarity. The anchor assembly 450 may be seen as having a plurality of discrete tissue anchors 456, e.g., barbs, hooks, projections, etc., each having a suture retaining end, e.g., an eyelet or opening 458 in a proximal end of the anchors 456. A suture member or wire 452 may be slidably connected to each anchor 456 through the openings 458 and through a cinching element 454, which may be configured to slide uni-directionally over the suture or wire 452 to approximate each of the anchors 456 towards one another. Each of the anchors 456 may be temporarily attached to the imaging hood 12 through a variety of methods. For instance, a pullwire or retaining wire may hold each of the anchors within a receiving ring around the circumference of the imaging hood 12. When the anchors 456 are released, the pullwire or retaining wire may be tensioned from its proximal end outside the patient body to thereby free the anchors 456 from the imaging hood 12.

One example for use of the anchor assembly 450 is shown in FIGS. 41A to 41D for closure of an opening or wound 460, e.g., patent foramen ovale (PFO). The deployment catheter 16 and imaging hood 12 may be delivered intravascularly into, e.g., a patient heart. As the imaging hood 12 is deployed into its expanded configuration, the imaging

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hood 12 may be positioned adjacent to the opening or wound 460, as shown in FIG. 41A. With the anchor assembly 450 positioned upon the expanded imaging hood 12, deployment catheter 16 may be directed to urge the contact edge of imaging hood 12 and anchor assembly 450 into the region surrounding the tissue opening 460, as shown in FIG. 41B. Once the anchor assembly 450 has been secured within the surrounding tissue, the anchors may be released from imaging hood 12 leaving the anchor assembly 450 and suture member 452 trailing from the anchors, as shown in FIG. 41C. The suture or wire member 452 may be tightened by pulling it proximally from outside the patient body to approximate the anchors of anchor assembly 450 towards one another in a purse-string manner to close the tissue opening 462, as shown in FIG. 41D. The cinching element 454 may also be pushed distally over the suture or wire member 452 to prevent the approximated anchor assembly 450 from loosening or widening.

Another example for an alternative use is shown in FIG. 42, where the deployment catheter 16 and deployed imaging hood 12 may be positioned within a patient body for drawing blood 472 into deployment catheter 16. The drawn blood 472 may be pumped through a dialysis unit 470 located externally of the patient body for filtering the drawn blood 472 and the filtered blood may be reintroduced back into the patient.

Yet another variation is shown in FIGS. 43A and 43B, which show a variation of the deployment catheter 480 having a first deployable hood 482 and a second deployable hood 484 positioned distal to the first hood 482. The deployment catheter 480 may also have a side-viewing imaging element 486 positioned between the first and second hoods 482, 484 along the length of the deployment catheter 480. In use, such a device may be introduced through a lumen 488 of a vessel VS, where one or both hoods 482, 484 may be expanded to gently contact the surrounding walls of vessel VS. Once hoods 482, 484 have been expanded, the clear imaging fluid may be pumped in the space defined between the hoods 482, 484 to displace any blood and to create an imaging space 490, as shown in FIG. 43B. With the clear fluid in-between hoods 482, 484, the imaging element 486 may be used to view the surrounding tissue surface contained between hoods 482, 484. Other instruments or tools may be passed through deployment catheter 480 and through one or more openings defined along the catheter 480 for additionally performing therapeutic procedures upon the vessel wall.

Another variation of a deployment catheter 500 which may be used for imaging tissue to the side of the instrument may be seen in FIGS. 44A to 45C. FIGS. 44A and 44B show side and end views of deployment catheter 500 having a side-imaging balloon 502 in an un-inflated low-profile configuration. A side-imaging element 504 may be positioned within a distal portion of the catheter 500 where the balloon 502 is disposed. When balloon 502 is inflated, it may expand radially to contact the surrounding tissue, but where the imaging element 504 is located, a visualization field 506 may be created by the balloon 502, as shown in the side, top, and end views of FIGS. 45A to 45C, respectively. The visualization field 506 may simply be a cavity or channel which is defined within the inflated balloon 502 such that the visualization element 504 is provided an image of the area within field 506 which is clear and unobstructed by balloon 502.

In use, deployment catheter 500 may be advanced intravascularly through vessel lumen 488 towards a lesion or tumor 508 to be visualized and/or treated. Upon reaching the

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lesion **508**, deployment catheter **500** may be positioned adjacently to the lesion **508** and balloon **502** may be inflated such that the lesion **508** is contained within the visualization field **506**. Once balloon **502** is fully inflated and in contact against the vessel wall, clear fluid may be pumped into visualization field **506** through deployment catheter **500** to displace any blood or opaque fluids from the field **506**, as shown in the side and end views of FIGS. **46A** and **46B**, respectively. The lesion **508** may then be visually inspected and treated by passing any number of instruments through deployment catheter **500** and into field **506**.

The applications of the disclosed invention discussed above are not limited to certain treatments or regions of the body, but may include any number of other treatments and areas of the body. Modification of the above-described methods and devices for carrying out the invention, and variations of aspects of the invention that are obvious to those of skill in the arts are intended to be within the scope of this disclosure. Moreover, various combinations of aspects between examples are also contemplated and are considered to be within the scope of this disclosure as well.

I claim:

1. A tissue imaging assembly comprising:

a first steerable catheter comprising a plurality of pull wires configured to control articulation of the first steerable catheter and a lumen, wherein the lumen opens through a distal face at a distal end of the first steerable catheter; and

a second steerable catheter deployable within the lumen of the first steerable catheter, the second steerable catheter comprising:

a lumen for delivering an elongate instrument out a distal region of the second steerable catheter; and an imager attached to the distal region of the second steerable catheter.

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2. The tissue imaging assembly of claim 1, wherein the elongate instrument comprises a guidewire.

3. The tissue imaging assembly of claim 1, wherein the elongate instrument comprises a needle.

4. The tissue imaging assembly of claim 1, wherein the elongate instrument comprises an electrode.

5. The tissue imaging assembly of claim 1, wherein the imager comprises an optical imaging fiber.

6. The tissue imaging assembly of claim 1, wherein the second steerable catheter further comprises an expandable hood.

7. A method comprising:

translating a second steerable catheter with respect to a first steerable catheter, the second steerable catheter supporting an imager attached to a distal region of the second steerable catheter and comprising a lumen for delivering an elongate instrument out the distal region of the second steerable catheter, the first steerable catheter comprising a plurality of pull wires configured to control articulation of the first steerable catheter and a lumen receiving the second steerable catheter, the lumen of the first steerable catheter opening through a distal face at a distal end of the first steerable catheter; inserting the elongate instrument through the lumen of the second steerable catheter; and

imaging a tissue region with the imager while the elongate instrument projects distally from the lumen of the second steerable catheter.

8. The method of claim 7, further comprising, expanding an imaging hood that is at a distal end of the second steerable catheter, the imaging hood being independently translatable from the first steerable catheter.

9. The method of claim 7, wherein the lumen of the second steerable catheter comprises a guidewire lumen and the elongate instrument comprises a guidewire.

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